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Department of Health and Human Services

Part 1. Overview Information

Participating Organization(s)

National Institutes of Health ([NIH \(http://www.nih.gov\)](http://www.nih.gov))

Components of Participating Organizations

Office of Strategic Coordination ([Common Fund \(http://commonfund.nih.gov/\)](http://commonfund.nih.gov/))

This Funding Opportunity Announcement (FOA) is developed as a Common Fund initiative (<https://commonfund.nih.gov/> (<https://commonfund.nih.gov/>)) through the Office of the NIH Director, Office of Strategic Coordination (<https://dpcpsi.nih.gov/> (<https://dpcpsi.nih.gov/>)). All NIH Institutes and Centers participate in Common Fund initiatives. The FOA will be administered by a trans-NIH team, which will be led by the Fogarty International Center ([FIC \(https://www.fic.nih.gov/\)](https://www.fic.nih.gov/)) on behalf of the NIH.

Funding Opportunity Title

Harnessing Data Science for Health Discovery and Innovation in Africa (DS-I Africa) Research Hubs (U54 Clinical Trial Optional)

Activity Code

U54 (https://grants.nih.gov/grants/funding/ac_search_results.htm?text_curr=u54&Search.x=0&Search.y=0&Search_Type=Activity)
Specialized Center- Cooperative Agreements

Announcement Type

New

Related Notices

- **July 29, 2020** - Notice of Technical Assistance Webinars for Harnessing Data Science for Health Discovery and Innovation in Africa (DS-I Africa) RFAs: RFA-RM-20-015, RFA-RM-20-016, RFA-RM-20-017, RFA-RM-20-018. See Notice [NOT-RM-20-024](https://grants.nih.gov/grants/guide/notice-files/NOT-RM-20-024.html) (<https://grants.nih.gov/grants/guide/notice-files/NOT-RM-20-024.html>).

Funding Opportunity Announcement (FOA) Number

RFA-RM-20-015

Companion Funding Opportunity

[RFA-RM-20-016 \(https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-016.html\)](https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-016.html) International Research Training Cooperative Agreements

[RFA-RM-20-017 \(https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-017.html\)](https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-017.html) Research Project Cooperative Agreements

[RFA-RM-20-018 \(https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-018.html\)](https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-018.html) High Impact Research and Research Infrastructure Cooperative Agreement Programs

Number of Applications

See [Section III. 3. Additional Information on Eligibility](#).

Catalog of Federal Domestic Assistance (CFDA) Number(s)

93.310

Funding Opportunity Purpose

This Funding Opportunity Announcement (FOA) seeks applications for Research Hubs as part of a broader initiative entitled [Harnessing Data Science for Health Discovery and Innovation in Africa \(DS-I Africa\)](#). The goal of DS-I Africa is to spur new health discoveries and catalyze innovation in healthcare, public health, and health research on the continent through application of data science.

DS-I Africa Research Hubs will be funded to advance data science health research and innovation in Africa and support new African and global partnerships that enhance the impact of data science health research. Hubs will engage government, industry, or other organizations, including diverse departments from academic institutions, that can partner to provide synergistic expertise, collaborate on research, and support data science linked to new technologies and solutions. Applicants must propose at least 2 distinct research projects and a system for soliciting, evaluating, and awarding smaller pilots projects, all focused on a central health theme that addresses priorities in an African country or region.

Applications must be submitted from African academic or other non-profit organizations. Proposed partnerships with organizations outside the applicant institution are required and can include all types of organizations in Africa, the U.S., or other countries.

Key Dates

Posted Date

July 24, 2020

Open Date (Earliest Submission Date)

November 9, 2020

Letter of Intent Due Date(s)

November 9, 2020

Application Due Date(s)

December 8, 2020

All applications are due by 5:00 PM local time of applicant organization. All [types of non-AIDS applications](#) allowed for this funding opportunity announcement are due on the listed date(s). Applicants are encouraged to apply early to allow adequate time to make any corrections to errors found in the application during the submission process by the due date.

AIDS Application Due Date(s)

February 8, 2021

All applications are due by 5:00 PM local time of applicant organization. All [types of AIDS and AIDS-related applications](#) allowed for this funding opportunity announcement are due on the listed date(s). Applicants are encouraged to apply early to allow adequate time to make any corrections to errors found in the application during the submission process by the due date.

Scientific Merit Review

April/May 2021 (<http://grants1.nih.gov/grants/funding/submissionschedule.htm#reviewandaward>)

Advisory Council Review

August 2021

Earliest Start Date

September 1, 2021

Expiration Date

February 9, 2021

Due Dates for E.O. 12372

Not Applicable

Required Application Instructions

It is critical that applicants follow the Multi-Project (M) Instructions in the [SF424 \(R&R\) Application Guide](http://grants.nih.gov/grants/guide/redirect.htm?id=12000) (<http://grants.nih.gov/grants/guide/redirect.htm?id=12000>), except where instructed to do otherwise (in this FOA or in a Notice from the [NIH Guide for Grants and Contracts](http://grants.nih.gov/grants/guide/) (<http://grants.nih.gov/grants/guide/>)). Conformance to all requirements (both in the Application Guide and the FOA) is required and strictly enforced. Applicants must read and follow all application instructions in the Application Guide as well as any program-specific instructions noted in [Section IV](#). When the program-specific instructions deviate from those in the Application Guide, follow the program-specific instructions. **Applications that do not comply with these instructions may be delayed or not accepted for review.**

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Part 2. Full Text of Announcement

Section I. Funding Opportunity Description

Background

Recent technological advances in Africa and around the world have enabled researchers to collect enormous volumes of data. From rural clinics to the most sophisticated genomics laboratories data are central to our ability to improve health, from delivering care to conducting biomedical research. The ability to fully extract useful knowledge from these data will lead to accelerated discoveries and innovations that can impact health in Africa and globally.

In the next decade, rapid advances in data science, including new approaches to the description, collection, storage, integration, and analysis of large, heterogeneous, structured and unstructured data sets, and new computational methods such as advanced deep learning, digital phenotypes, machine learning and artificial intelligence, and 3D imaging are expected to transform biomedical and

behavioral research and lead to improved health for individuals and populations. Traditional datasets (e.g. national health systems, surveillance, surveys) are becoming deeper and richer while new sources of data based on new technologies and sensors (e.g. social media, geospatial data, mobile phones, wearables, electronic medical records, bioimaging, and genomics) are emerging that may be of greatest value when linked to data science. Progress in development of new large data sets and advanced methods for mining them underpin advances in diagnostics, technology development, and the potential for precision public health. The Harnessing Data Science for Health Discovery and Innovation in Africa (DS-I Africa) program is designed to spur discoveries and innovations through application of data science in the African context that will ultimately promote significant improvements in health for African individuals, communities, and populations. We define data science as the interdisciplinary field of inquiry in which quantitative and analytical approaches, processes, and systems are developed and used to extract knowledge and insights from increasingly large and/or complex sets of data ([NIH Strategic Plan for Data Science \(https://datascience.nih.gov/strategicplan\)](https://datascience.nih.gov/strategicplan)).

Decades of infrastructure building and training in Africa have provided compelling opportunities for research to address the disproportionate share of the global burden of disease that Africa bears. Data science has the potential to significantly impact both quantitative and qualitative research and health on the continent. By leveraging existing digital infrastructure, new solutions can be developed that are relevant, affordable, acceptable, and scalable. For example, extensive mobile phone coverage in Africa has led to major innovations in banking and agriculture and similarly has the potential to rapidly advance health care delivery systems, bringing the clinic to the patient through point of care technologies and self-management systems, with applications to rural and underserved populations worldwide. Additionally, this initiative is synergistic with and leverages the substantial investments NIH, the President's Emergency Plan for AIDS Relief (PEPFAR), and other US government entities have already made in research and research training in Africa. NIH funds more than 1,600 active grants working in Africa and has catalyzed well-developed clinical and academic research networks through prior investments by the NIH Common Fund and its partners in the Medical Education Partnership Initiatives ([MEPI \(https://www.fic.nih.gov/Programs/Pages/medical-education-africa.aspx\)](https://www.fic.nih.gov/Programs/Pages/medical-education-africa.aspx)) and the Human Health and Heredity in Africa ([H3Africa \(https://h3africa.org/\)](https://h3africa.org/)) consortium.

DS-I Africa Research Hubs will serve as a core component within the larger DS-I Africa program, which will also consist of 1) data science and innovation training programs, 2) research on the ethical, legal, and social implications (ELSI) central to data science health research and innovation in Africa, and 3) an open data science platform and coordinating center. As a group, awardees will constitute a collaborative consortium, made up of all programs funded through the broader DS-I Africa program, and will work collectively toward strategic objectives designated by the consortium, beyond the individual research objectives of each award. Representatives from each DS-I Africa Research Hub will be expected to participate fully in the DS-I Africa Consortium and cross-consortium technical and education working groups.

Research Objectives

A. Scientific Scope

The overarching goal of the DS-I Africa program is to advance data science and related innovations in Africa through creation of a multisectoral and multidisciplinary ecosystem that can begin to provide local solutions to countries most immediate biomedical and public health problems through advances in research. It is expected that the DS-I Africa Research Hubs will become recognized centers of excellence in various fields of data science that can be regional and continental resources. Additionally, these hubs will have the ability to advance population-relevant, affordable, acceptable, and scalable data science solutions and, in the long-term, demonstrate the ability of these solutions to improve health in Africa.

DS-I Africa Research Hubs should be interdisciplinary (biomedical science, computer science, engineering, social and behavioral science, and other fields) and focus on a critical health research area to be addressed through data science approaches. This critical health research area should be of importance to the country or region of focus, and may be cross-cutting, disease-specific, or data-specific. *Hubs must include at least two partners outside the applicant institution* (one of which must be a non-academic African organization). Applicants are highly encouraged to propose methods to form, maintain, and/or strengthen African governmental and private partnerships (including industry, NGOs, and related organizations globally), in addition to academic partners from other countries globally, to provide synergistic expertise, collaborate on research, and support data science linked to new technologies. It is expected that these partnerships will expand over the course of the award and collectively support the various aspects of the research supported by the Research Hub.

Together, the Research Hub collaborations must provide the complete capacity needed to carry out research on a health topic important to Africa. There are many areas of data science that may be engaged by the Research Hubs and projects will be expected to employ a variety of approaches through the research proposed. Projects and Cores may collect and utilize a wide range of data types and sources, including but not limited to clinical, public health, genomic, imaging, geospatial, mobile, and environmental data. Hubs should leverage existing, African-based information and communication technology infrastructure, such as cell coverage and mobile phones, satellites, high-performance computing infrastructure and other digital information systems.

The proposed Research Hub should include several demonstrated competencies, as appropriate to the project proposed, to meet the overarching goals of the DS-I Africa program, which include leveraging open data science, biomedical domain knowledge, technology, capacity building, and community/stakeholder engagement to advance health discovery and innovation in Africa. Each DS-I Africa Research Hub will be expected to include the following components:

- Research Projects (*required, minimum of two*)
- Administrative Core (*required, maximum of one*)
- Data Management and Analysis Core (*required, maximum of one*)
- Support Cores (*optional*)

Research Projects: Applicants must propose at least two research projects. One research project must be solution-oriented through the development or adaptation of a product (software or hardware) or new technology that is either needed for the collection of new data types or translates initial data science research into a new health innovation. Other research projects proposed will be up to the applicants to define. While new data can be collected where necessary, projects should primarily focus on use of pre-existing data. In the long-term, Research Hubs should demonstrate through these projects the potential for data science-driven research products to impact health, healthcare, or health research and translate beyond the research lab. See "Specific Areas of Research Interest" below for further information about applicable areas of research.

Administrative Core: This core leads the Research Hub, maintains the organization of the Hub and its external collaborations, and ensures that the projects and cores are being supported to achieve the Hub's goals. The Administrative Core facilitates and coordinates communication of Hub results, accomplishments, and implications of the Hub's research to stakeholders, including appropriate NIH staff, the DS-I Africa Coordinating Center, DS-I Africa affiliated programs, and other stakeholders in a context that makes the research applicable to the target audience. The Administrative Core has the critical role in assisting project/core leaders in translating their research outcomes and activities to appropriate audiences (including industry partners for applicable commercialization of Hub-related innovations); ensuring the accurate and timely use of products and sharing of findings; and advancing appropriate ethical, legal, and social implications research tied to research projects and overall Hub activities. This core also has the primary role of initiating partnerships, arranging appropriate subawards, determining and distributing award funds, etc. as partnerships expand over the course of the award.

Additionally, a key component of each DS-I Africa Research Hub must involve a mechanism to provide sub-award support for pilot projects that have significant potential to address developing needs and opportunities while being shorter term and varying throughout the period of award. These pilot projects will allow the grantees to take advantage of new directions stimulated by their projects, form new partnerships, provide support to early-stage investigators to undertake data science-related research and innovation, and support relevant cross-consortium projects. Beyond the administrative oversight activities of an NIH-sponsored Administrative Core, it is expected that the DS-I Research Hub Administrative Core will establish and manage a solicitation, review, and selection process for collaborative sub-award pilot projects aligned with the Research Hub overall focus and milestones, including the distribution of yearly sub-award funds. *Hubs are required to fund two to five sub-award projects per year beginning in Year 2 of the research funding cycle*, with at least one yearly sub-award project from outside the existing Hub organizational structure, in keeping with the overarching DS-I Africa goal of sustaining and expanding vital partnerships. Hubs should also encourage junior investigator-led sub-award pilot projects from within and outside the U54 awardee institution, as well as serve as potential research sites for trainees associated with the DS-I Africa Research Training Program.

The Administrative Core should also lead or propose how the Research Hub will interact with other DS-I Africa Research Hubs, Training Programs, ELSI Projects, Open Data Science Platform, and Coordinating Center for advancing aligned research (as applicable), strengthened research capacity building efforts, informed community engagement and development of relevant ethical and legal frameworks, and open data sharing. These efforts can be led directly through the Administrative Core or through an optional *Support Core*.

Data Management and Analysis Core: Data represent important research products and attention is needed to provide stewardship of data and treat data as an asset. To support NIH data sharing policies (<https://grants.nih.gov/policy/sharing.htm>) and promote best practices, all DS-I Africa Research Hub applicants are required to have a Data Management and Analysis Core (DMAC). The primary purpose of the DMAC is to support the management, integration, and interoperability (as appropriate) of data assets across the Hub, including the Hub research projects and all pilot projects. The DMAC will support management and analysis of both pre-existing data and new data produced through the Hub, as well as the development of tools for data assessment. Developing creative approaches that foster the development or strengthening of artificial-intelligence ready data sets (AI-Ready data sets) is desired. The DMAC should also establish, coordinate, and monitor processes for data analysis and work closely with project/core leaders to ensure high data quality throughout the entire lifecycle of the data. The DMAC also has a coordinating role to work with project/core leaders to identify opportunities for integrating project/core-generated data with other existing datasets, as appropriate. Importantly, the DMAC should also facilitate collaboration with the DS-I Africa Open Data Science Platform for applicable alignment, management, and harmonization of data across the DS-I Africa Consortium.

Research Hub Cores (both required Cores and optional Support Cores) should support and strengthen the efforts of the Research Projects and Pilot Projects, as well as further enable integration with the broader DS-I Africa Consortium efforts.

Potential Areas of Research

While the specific area(s) a Research Hub application proposes to address will be determined by the applicant, potential topics that could be considered include, but are not limited to:

- multi-omics computational research (e.g., genomics, epigenomics, transcriptomics, epitranscriptomics, proteomics, pharmacogenetics, and biomarkers), including applied research efforts that work toward the development of polygenic risk scores for defined African populations;
- development and use of shared platforms and leveraging of large health datasets that enable discovery of emergent multi-morbid patterns as well as longitudinal risk;
- computational approaches to mine electronic health records, data from digital health devices and social media to generate evidence to enhance findings from clinical trials;
- development of algorithms for extraction of features from clinical reports to reduce human resource needs and develop pattern recognition in population data to enable prediction of treatment outcomes;
- development and leveraging of novel machine learning/artificial intelligence/computer vision tools for data analytics, computer-assisted detection and diagnosis or treatment planning that translate affordable point of need technologies and mobile medical decision support;
- computational approaches that improve the remote transmission and clinical utility of imaging technologies to rural hospitals and clinics;
- integration of sensor technologies with satellite and mobile data, demographics, exposure biomarkers, and other data for exposure assessment and intervention;
- development or expansion of informatic tools and statistical methodologies to analyze multi-dimensional environmental exposure data, gene-environment interaction (GxE) data, and related complex data;
- computational approaches to improve understanding and surveillance of risk factors, diseases, and/or disease susceptibility;
- data-driven and computational approaches for examining developmental trajectories of health and disease (including disparities, life-course patterns, and/or generational differences) in diverse populations in Africa;
- computational approaches and automated, real-time data sharing practices to accelerate the discovery and development of new therapeutics, diagnostics and vaccines;
- data science technologies and approaches for behavioral health interventions that leverage novel data sources, new and emerging sensors, and emerging machine learning approaches;
- computational tools and technologies for the development of new approaches to provide formal and informal care in distinct regions of Africa that are experiencing the dual burden of communicable diseases and non-communicable diseases;
- computational technologies and approaches to improve the capacity of local, national, or regional public health planning and public health systems in Africa;
- development of data visualization and analysis tools to accelerate knowledge extraction from data that is relevant at the local, country and regional level in Africa and globally;
- modelling applications to predict disease outbreaks, pathogen exposure, disease dynamics, and susceptibility; and
- modelling of markers for Precision Public Health and the use of big data to predict public health risk and customize treatments for more specific and homogeneous subpopulations, among many others.

Applicants are encouraged to consider the unique opportunities for African research, including:

- extensive genomic diversity, migration, and admixture;
- diseases and health conditions prevalent on the African continent;
- environmental exposures and health-related conditions; and
- local health priorities.

All applications proposing research to improve the understanding of HIV/AIDS prevention, diagnosis, treatment, cure and related complications will be considered in accordance with the NIH AIDS Research Priorities document found at the following link (<https://grants.nih.gov/grants/guide/notice-files/NOT-OD-15-137.html> (<https://grants.nih.gov/grants/guide/notice-files/NOT-OD-15-137.html>)).

Applications submitted in response to this FOA may propose research that is cross-cutting or more specifically focused on a disease or health area that falls within the broad areas of data science for health discovery and innovation. Applications may fall within the mission of any NIH institute, center, or office and additional information is provided regarding specific areas of interest of members of the DS-I Africa working group on the DS-I Africa program website: <https://commonfund.nih.gov/AfricaData> (<https://commonfund.nih.gov/AfricaData>).

B. Additional Objectives of DS-I Africa Program

As noted above and in the companion FOAs, there are several specific objectives that the DS-I Africa Program is trying to achieve. Successful applicants will become members of the DS-I Africa Consortium, and will be expected to adhere to these policies.

DS-I Africa Consortium Participation: DS-I Africa will be organized as a research consortium that brings participants together in a highly collaborative and synergistic effort. The DS-I Africa Consortium will include all participants of research, training, ELSI, and open data science platform/coordinating center programs funded through DS-I Africa, as well as responsible NIH staff. Groups funded under this program will be expected to participate in the DS-I Africa Consortium and to collaborate effectively with each other to maximize the chances of overall success of the program. Each funded applicant is expected to participate directly or via proxy in Consortium Working Groups that establish rules, guidelines, and resources for the Consortium or that develop cross-consortium research or related projects. Each project is expected to comply with applicable consortium policies and procedures. In addition, the PD/PI(s) will be active members of the DS-I Africa Steering Committee, which will meet regularly by teleconference. The DS-I Africa Consortium will generally hold two meetings per year, usually in Africa at Consortium member sites.

DS-I Africa Data Sharing Policies: Data sharing is an increasingly important aspect of contemporary biomedical research, and the Consortium will develop a Data Sharing and Access Policy within year 1 of the award that applies to data and resources funded with NIH support. The policy will address the data sharing needs of the Consortium while safeguarding the privacy of research participants and protecting confidential and proprietary data and resources. One of the roles of the associated DS-I Africa Open Data Science Platform is to facilitate data and resource access and sharing, and all DS-I Africa awardees will be expected to work closely with the Open Data Science Platform team to facilitate appropriate alignment of activities. The NIH Office of Data Science Strategy ([ODSS \(https://datascience.nih.gov/\)](https://datascience.nih.gov/)), guided by the [NIH Strategic Plan for Data Science \(https://datascience.nih.gov/strategicplan\)](https://datascience.nih.gov/strategicplan), is committed to ensuring that data-science activities and products supported by the NIH are Findable, Assessable, Interoperable, and Reusable (FAIR). Data, analytical, educational, and other resources generated by a DS-I Africa awardee are expected to conform to the [FAIR Principles \(https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4792175/\)](https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4792175/), where possible. Additionally, guidance may be developed by the Consortium to inform data acquisition, collection and curation, including appropriate consent for data sharing, as appropriate for the Consortium member projects and their relevant standards or schemas.

Partnerships and Collaborations: One of the major goals of the DS-I Africa program is to enhance the impact of data science on health in Africa and spur innovation through support of new African and global partnerships. These partnerships are expected to contribute to the sustainability of African data science programs and the advancement of product development. Awardees are expected to maintain and strengthen existing partnerships that are proposed in applications and engage new partners throughout the duration of the award funding. Collaboration within the DS-I Africa consortium will also be facilitated through the DS-I Africa Coordinating Center. The DS-I Africa Research Hubs will have the opportunity to develop cross-consortium research projects enabled by the DS-I Africa Open Data Science Platform. DS-I Africa Research Training Programs will have the opportunity to collaborate with the DS-I Africa Research Hubs through mentored training experiences for trainees. Finally, DS-I Africa ELSI awardees will collaborate with the Research Hubs to explore and address issues that are specifically relevant to the data science projects within the consortium.

Community Engagement: NIH recognizes ongoing [community engagement \(https://datascience.nih.gov/community-engagement\)](https://datascience.nih.gov/community-engagement) and the building of trust relationships with research participants as an essential feature of ethical biomedical and population-based data science research involving human subjects. DS-I Africa awardees are expected to incorporate relevant community engagement activities into data science research and research training activities. Community engagement can include a variety of activities, including those that empower communities to contribute to and become partners in data science research and activities that engage diverse stakeholders to learn about and contribute to DS-I Africa efforts in data science.

Data Science Capacity in Africa: Another major objective for the DS-I Africa program is to increase capacity at African institutions to use and develop data science approaches in health research and innovation. The DS-I Africa Research Training Programs will build individual and institutional data science capacity through development of new data science curriculum and training of data science research leaders in master's, PhD, post-doctoral, and faculty development programs. Research in DS-I Africa Research Hubs and DS-I Africa ELSI awards should provide a variety of training and career enhancement opportunities to students, postdoctoral researchers, and young investigators in diverse disciplines through mentored research experiences. Additionally, Research Hub pilot projects should provide opportunities for early career data scientists to initiate small research projects that could launch their careers. The DS-I Africa Open Data Science Platform and Coordinating Center will enhance activities conducted by the other awardees by developing short courses relevant to data science and innovation that fill training gaps in the consortium and have potential to reach the broader African research community. Long-term sustainability and institutional/governmental commitments to research education programs and independent career opportunities are objectives of the DS-I Africa program.

Technical Assistance and DS-I Africa Virtual Symposium

All applicants are strongly encouraged to contact NIH Staff early in the application process to discuss the alignment of their application with the goals of this FOA, and with the DS-I Africa program. Technical Assistance Sessions and teleconferences will be held for potential applicants to this FOA and companion FOAs. NIH staff will be available to answer questions related to this FOA. Location, time, date, and connection information will be announced in an NIH Guide Notice and will be posted on the DS-I Africa website: <https://commonfund.nih.gov/AfricaData> (<https://commonfund.nih.gov/AfricaData>). During the Technical Assistance Sessions, NIH staff will present an overview of these FOAs and answer questions from prospective applicants. Additionally, these technical assistance sessions will be part of a larger DS-I Africa Virtual Symposium, which will examine the state of data science in Africa and facilitate networking across disciplines, sectors, and geographies to promote data science for health in Africa. Registration and other details will be posted on the Virtual Symposium website: <https://commonfund.nih.gov/africadatasymposium> (<https://commonfund.nih.gov/africadatasymposium>). All virtual events are free and open to all prospective applicants, but participation is not a prerequisite to applying.

See [Section VIII. Other Information](#) for award authorities and regulations.

Section II. Award Information

Funding Instrument

Cooperative Agreement: A support mechanism used when there will be substantial Federal scientific or programmatic involvement. Substantial involvement means that, after award, NIH scientific or program staff will assist, guide, coordinate, or participate in project activities. See Section VI.2 for additional information about the substantial involvement for this FOA.

Application Types Allowed

New

The [OER Glossary \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11116\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11116) and the SF424 (R&R) Application Guide provide details on these application types. Only those application types listed here are allowed for this FOA.

Clinical Trial?

Optional: Accepting applications that either propose or do not propose clinical trial(s)

[Need help determining whether you are doing a clinical trial? \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=82370\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=82370)

Funds Available and Anticipated Number of Awards

The NIH Common Fund intends to commit an estimated total of \$32,500,000 over five years to fund up to 5 awards.

The number of awards is contingent upon NIH appropriations and the submission of a sufficient number of meritorious applications.

Award Budget

The requested budget needs to reflect the actual needs of the proposed project. DS-I Africa Research Hub applications are limited to \$1,300,000 per year in total costs.

Award Project Period

The scope of the proposed DS-I Africa Research Hub should determine the project period. The maximum project period is 5 years.

NIH grants policies as described in the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11120\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11120) will apply to the applications submitted and awards made from this FOA.

Section III. Eligibility Information

1. Eligible Applicants

Eligible Organizations

Other

Non-domestic (non-U.S.) Entities (Foreign Institutions)

- African Public/State Controlled Institutions of Higher Education
- African Private Institutions of Higher Education
- African Nonprofits Other Than Institutions of Higher Education

African applicants may propose partners from all types of organizations in Africa, the U.S., or other countries, including for-profit and government entities.

Foreign Institutions

Non-domestic (non-U.S.) Entities (Foreign Institutions) **are** eligible to apply.

Non-domestic (non-U.S.) components of U.S. Organizations **are not** eligible to apply.

Foreign components, as [defined in the NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11118\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11118), **are** allowed.

Required Registrations

Applicant Organizations

Applicant organizations must complete and maintain the following registrations as described in the SF 424 (R&R) Application Guide to be eligible to apply for or receive an award. All registrations must be completed prior to the application being submitted. Registration can take 6 weeks or more, so applicants should begin the registration process as soon as possible. The [NIH Policy on Late Submission of Grant Applications \(//grants.nih.gov/grants/guide/notice-files/NOT-OD-15-039.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-15-039.html) states that failure to complete registrations in advance of a due date is not a valid reason for a late submission.

- [Dun and Bradstreet Universal Numbering System \(DUNS\) \(http://fedgov.dnb.com/webform\)](http://fedgov.dnb.com/webform) - All registrations require that applicants be issued a DUNS number. After obtaining a DUNS number, applicants can begin both SAM and eRA Commons registrations. The same DUNS number must be used for all registrations, as well as on the grant application.
- [System for Award Management \(SAM\) \(https://www.sam.gov/portal/public/SAM/\)](https://www.sam.gov/portal/public/SAM/) Applicants must complete and maintain an active registration, **which requires renewal at least annually**. The renewal process may require as much time as the initial registration. SAM registration includes the assignment of a Commercial and Government Entity (CAGE) Code for domestic organizations which have not already been assigned a CAGE Code.
- [NATO Commercial and Government Entity \(NCAGE\) Code \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11176\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11176) Foreign organizations must obtain an NCAGE code (in lieu of a CAGE code) in order to register in SAM.
- [eRA Commons \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11123\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11123) - Applicants must have an active DUNS number to register in eRA Commons. Organizations can register with the eRA Commons as they are working through their SAM or Grants.gov registration, but all registrations must be in place by time of submission. eRA Commons requires organizations to identify at least one Signing Official (SO) and at least one Program Director/Principal Investigator (PD/PI) account in order to submit an application.
- [Grants.gov \(//grants.nih.gov/grants/guide/url_redirect.htm?id=82300\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=82300) Applicants must have an active DUNS number and SAM registration in order to complete the Grants.gov registration.

Program Directors/Principal Investigators (PD(s)/PI(s))

All PD(s)/PI(s) must have an eRA Commons account. PD(s)/PI(s) should work with their organizational officials to either create a new account or to affiliate their existing account with the applicant organization in eRA Commons. If the PD/PI is also the organizational Signing Official, they must have two distinct eRA Commons accounts, one for each role. Obtaining an eRA Commons account can take up to 2 weeks.

Eligible Individuals (Program Director/Principal Investigator)

Any individual(s) with the skills, knowledge, and resources necessary to carry out the proposed research as the Program Director(s)/Principal Investigator(s) (PD(s)/PI(s)) is invited to work with his/her organization to develop an application for support. Individuals from underrepresented racial and ethnic groups as well as individuals with disabilities are always encouraged to apply for NIH support.

For institutions/organizations proposing multiple PDs/PIs, visit the Multiple Program Director/Principal Investigator Policy and submission details in the Senior/Key Person Profile (Expanded) Component of the SF424 (R&R) Application Guide.

The use of the multiple PD/PI (MPI) administrative structure is *strongly encouraged* for this program. The PD/PI(s) should be established investigators in their fields and capable of providing both administrative and scientific leadership to the development and implementation of the proposed program. The contact PD/PI will be designated as the Research Hub Director. Research Hub Deputy Director(s) can also be named at the discretion of the applicants. The PD/PI(s) should have research experience in Africa, particularly

in the country where the proposed research program will be established. The PD/PI(s) will be responsible for the overall direction, management, administration, and evaluation of the program. The PD/PI(s) will be expected to monitor and assess the program and submit all documents and reports as required. The PD/PI(s) have responsibility for the program and for appointing members of a Scientific Advisory Board, including using their recommendations to modify the overall direction, management, administration, and evaluation of the program.

The contact PD/PI (i.e., the Research Hub Director) must be affiliated with the African institution submitting the application where the proposed research program will be established and must have citizenship in an African country. Other PD/PI(s) with relevant expertise from partner institutions in Africa, U.S. or other country institutions may be proposed. Multiple PD/PI(s) should have a documented history of collaboration relevant to the proposed program or clear rationale for a new collaboration to meet the strategic goals of the proposed Research Hub.

2. Cost Sharing

This FOA does not require cost sharing as defined in the [NIH Grants Policy Statement](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11126). ([//grants.nih.gov/grants/guide/url_redirect.htm?id=11126](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11126))

3. Additional Information on Eligibility

Number of Applications

Applicant organizations may submit more than one application, provided that each application is scientifically distinct.

The NIH will not accept duplicate or highly overlapping applications under review at the same time. This means that the NIH will not accept:

- A new (A0) application that is submitted before issuance of the summary statement from the review of an overlapping new (A0) or resubmission (A1) application.
- A resubmission (A1) application that is submitted before issuance of the summary statement from the review of the previous new (A0) application.
- An application that has substantial overlap with another application pending appeal of initial peer review (see [NOT-OD-11-101](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-11-101.html) ([//grants.nih.gov/grants/guide/notice-files/NOT-OD-11-101.html](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-11-101.html))).

Section IV. Application and Submission Information

1. Requesting an Application Package

The application forms package specific to this opportunity must be accessed through ASSIST or an institutional system-to-system solution. A button to apply using ASSIST is available in [Part 1](#) of this FOA. See your administrative office for instructions if you plan to use an institutional system-to-system solution.

2. Content and Form of Application Submission

It is critical that applicants follow the Multi-Project (M) Instructions in the [SF424 \(R&R\) Application Guide](https://grants.nih.gov/grants/guide/url_redirect.htm?id=12000) ([//grants.nih.gov/grants/guide/url_redirect.htm?id=12000](https://grants.nih.gov/grants/guide/url_redirect.htm?id=12000)), except where instructed in this funding opportunity announcement to do otherwise and where instructions in the Application Guide are directly related to the Grants.gov downloadable forms currently used with most NIH opportunities. Conformance to the requirements in the Application Guide is required and strictly enforced. Applications that are out of compliance with these instructions may be delayed or not accepted for review.

Letter of Intent

Although a letter of intent is not required, is not binding, and does not enter into the review of a subsequent application, the information that it contains allows IC staff to estimate the potential review workload and plan the review.

By the date listed in [Part 1. Overview Information](#), prospective applicants are asked to submit a letter of intent that includes the following information:

- Descriptive title of proposed activity
- Name(s), address(es), and telephone number(s) of the PD(s)/PI(s)
- Names of other key personnel
- Participating institution(s)
- Number and title of this funding opportunity

The letter of intent should be sent to:

Brad Newsome, Ph.D.

Telephone: 301-480-8389

Fax: 301-402-2173

Email: brad.newsome@nih.gov (<mailto:brad.newsome@nih.gov>)

Page Limitations

Available Component Types	Research Strategy/Program Plan Page Limits
Overall	12
Core (use for Administrative Core, Data Management and Analysis Core, and optional Support Cores)	6
Project (use for Research Projects)	12

Additional page limits described in the SF424 Application Guide and the [Table of Page Limits](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11133) (https://grants.nih.gov/grants/guide/url_redirect.htm?id=11133) must be followed.

Instructions for the Submission of Multi-Component Applications

The following section supplements the instructions found in the SF424 (R&R) Application Guide, and should be used for preparing a multi-component application.

The application should consist of the following components:

- Overall: required
- Administrative Core: required, maximum of 1
- Data Management and Analysis Core: required, maximum of 1
- Support Cores: optional
- Research Projects: required, minimum of 2

Overall Component

When preparing your application, use Component Type Overall .

All instructions in the SF424 (R&R) Application Guide must be followed, with the following additional instructions, as noted.

SF424 (R&R) Cover (Overall)

Complete entire form.

PHS 398 Cover Page Supplement (Overall)

Note: Human Embryonic Stem Cell lines from other components should be repeated in cell line table in Overall component.

Research & Related Other Project Information (Overall)

Follow standard instructions.

- **Project Summary/Abstract:** For the Overall Component, applicants must include a Project Summary/Abstract that briefly describes the problem(s) being addressed by the Overall Research Hub, how the Hub plans to solve target problem(s) related to data science for health discovery and innovation in Africa, and the major participating countries proposed. As opposed to a generic description of the Hub components, the Project Summary should provide an overall indication of the proposed science and what the Hub proposes to accomplish. For example, the summary should describe which health-related questions are being evaluated, study sites/populations, research approaches, data science methodologies, available infrastructure and data sets, new data required to answer research questions, technology development, etc. In addition, the summary should highlight aspects of the application that are particularly innovative or paradigm-shifting. The primary activities of the cores should be integrated into the summary. The benefits of the Hub (in terms of improving healthcare or public health through data science and innovation) should also be included. Note: the project summary should be written in plain language for a diverse set of reviewers while providing a depth of understanding of the science and activities of the Research Hub.
- **Project Narrative:** It should be clear in the "Project Narrative" (i.e., the "public health relevance" statement) the relevance of the Hub's activities to public health, diverse stakeholders, and the DS-I Africa Program broadly.
- **Other Attachments:** The following "Other Attachments" must be included with the overall component to aid in the review of applications. The filename provided for each attachment will be the name used for the bookmark in the application image.
- **Hub Organizational Structure.** Applicants must include a diagram of the organizational structure of the Hub. This diagram should demonstrate how the interactions between the different Hub components achieve integration and the overarching goals of the Hub. The diagram should be provided as a pdf file titled "Hub Organizational Structure".

Project/Performance Site Location(s) (Overall)

Enter primary site only.

A summary of Project/Performance Sites in the Overall section of the assembled application image in eRA Commons compiled from data collected in the other components will be generated upon submission.

Research & Related Senior/Key Person Profile (Overall)

Include only the Project Director/Principal Investigator (PD/PI) and any multi-PDs/PIs (if applicable to this FOA) for the entire application.

A summary of Senior/Key Persons followed by their Biographical Sketches in the Overall section of the assembled application image in eRA Commons will be generated upon submission.

Budget (Overall)

The only budget information included in the Overall component is the Estimated Project Funding section of the SF424 (R&R) Cover.

A budget summary in the Overall section of the assembled application image in eRA Commons compiled from detailed budget data collected in the other components will be generated upon submission.

PHS 398 Research Plan (Overall)

Specific Aims: Applicants must provide Specific Aims for the Overall Research Hub describing the objectives and goals of the Research Hub, as they relate to the overarching goals of the DS-I Africa Program. It should be clear how the components of the Hub will interact to solve the target problem(s), and the role of each project and core in contributing to resolving the problem(s) stated in the application. This section should provide greater scientific detail and approaches than the overall Project Summary/Abstract. In addition, each specific aim should be supported by a brief description of how that aim will be accomplished through interactions of projects and cores.

Research Strategy: The Research Strategy section for the Overall Research Hub should highlight the significance of the Hub's overall goals relative to the DS-I Africa program goals, the group expertise of the research team to accomplish the goals (without duplicating information in biosketches), the roles of team members and organizations outside the applicant institution, the innovation of the science, and the strengths of the experimental approaches and the research environment. Hubs should detail the integration between projects and cores, with a focus on how the Overall Hub is greater than the sum of its parts (i.e., component sub-programs). As such, the following subsections are recommended: Significance and Relevance to the DS-I Africa Program; Research Team and Environment; Innovation; Approach; Sustainability; and Hub Integration.

Significance and Relevance to the DS-I Africa Program: Applicants should include an overview of the Hub that clearly describes the problem(s) being addressed by the Overall Hub, how the components of the Hub interact to solve the target problem(s), and the role of each project and core in contributing to resolving the problem(s) related to data science for health discovery and innovation in Africa. The problem being addressed could be a discreet but critical gap in the larger context of a complex issue, or it may answer a discrete question that, if left unanswered, would create an impediment to effectively protecting public health. This section should also include background information and a description of how the Hub's research addresses the broad objectives of the DS-I Africa program (see Part 2. Section I. Additional Objective of DS-I Africa Program for more information).

Research Team and Environment: Applicants should introduce the leadership and key personnel of the Hub, highlighting the breadth of disciplines interacting among the research projects and cores. Applicants should report how complementary and integrated expertise among research project and core leaders accomplishes the goals of the Hub. Applicants should report the successes and accomplishments of the team as a whole in advancing their respective field - particularly with regard to integration between disciplines. Applicants should include a brief description of plans for the Administrative Core to facilitate the overall goals of the Research Hub and maintain organization of the Hub, particularly with respect to collaborations that are not located at the PD/PI(s) home institution. Applicants should introduce and justify any optional support cores that the applicant wishes to include in the DS-I Africa Research Hub. Applicants should also highlight how the institutional environment supports the Hub's problem-solving goals, provides institutional support including unique data science capabilities, enhances interdepartmental/interinstitutional cooperation needed to carry out the multi-disciplinary activities, and enables a move toward sustainability during and beyond the period of funding requested. If proposed, include a description of in-kind support being provided by the institutions or partners.

Innovation: A description of the innovative aspects of the Research Hub should be included. Applicants should describe how the proposed data science-driven approaches lead toward a novel solution to critical health barriers or advance understanding of critical health challenges facing Africa. Applicants should highlight how related theoretical concepts, approaches or

methodologies, available data sets and infrastructure, and/or existing interventions (including novel adaptations of technologies from one field to another) can be leveraged to answer associated critical research questions.

Approach: The methods, technologies, and innovative approaches that will be employed by the DS-I Africa Research Hub should be identified and the impact of those methods/technologies/approaches on the future of African science, public health, and/or medicine should be described; more detailed descriptions can be provided in the individual project and core sections. Challenges and problems that may be encountered should also be addressed and suggestions should be provided as to how they may be solved or overcome. Issues of data acquisition, data management, data storage, and analytical capability that the Research Hub will confront should be concisely discussed. Further details of the Research Hub's data science and innovation facilitation activities should be discussed in the Data Management and Analysis Core Research Strategy as well as embedded, as appropriate, within the individual research projects and pilot projects.

Overall Research Hub Milestones must be included addressing how the key goals of the Research Hub will be achieved. A milestone is defined as a scheduled event in the project timeline that signifies the completion of a major project stage or activity. Milestones must be performance-based to enhance the likelihood that the project will be completed on-time and on-budget. In the event of an award, the PD/PI(s) and NIH staff will negotiate the final list of milestones for each year of support. If at any time the project fails to make progress toward meeting milestones (e.g., developing a final protocol and/or manual of procedures including a detailed description of study procedures and process details; completing training of study staff; etc.), the NIH may consider ending support and negotiating an orderly close-out of the award. An associated Overall Hub timeline should provide further information for the completion of project/core activities associated with each milestone.

Sustainability: The DS-I Africa program should enable African scientists to demonstrate their world-class skills through the establishment of cutting-edge research programs that will lead to more publications, digital assets, and other evidence of productivity, thereby increasing their opportunities for future funding through competitive grant processes, as well as through increased partnership with and potential support from national governments and private sources. Dedicated funds for DS-I Africa by the NIH Common Fund will provide 5 to 10 years of support. It is expected that outcomes associated with the Research Hub will be completed within 10 years and will move toward commercialization, where appropriate. Beyond that, any support for continuation of the research programs and research careers initiated under the auspices of DS-I Africa will be dependent on other funding. Therefore, applicants need to discuss the issue of future sustainability of their Research Hub activities beyond the DS-I Africa program, including potential for commercialization of research products or other private sector support, where applicable, and how these activities will contribute to success in continuing the broad efforts of its participants. This section of the application should refer to the letters of Institutional and National commitments to provide an overview of the long-term prospects for sustained research.

Research Hub Integration: This section should include a description of the multidisciplinary, interdisciplinary, and multisectoral nature of the Hub, the interactions between the projects and cores, and potential cross-Hub and cross-consortium data analyses; how each project and core contributes to the Hub's theme and overall success; and how the Research Hub will leverage new and pre-existing data and African digital infrastructure to address relevant health challenges. Applicants should provide evidence of integration of the Administrative and Data Management and Analysis Cores with the Research Projects and Pilot Projects (and Support Cores, if applicable). The applicant should also refer to data management activities (i.e., Overall Resource Sharing Plan; DMAC components; and Project/Core Resource Sharing Plans) for evidence of integration between projects and cores and how data sharing enhances not only the impact of the Hub's research and activities but also those of the broader DS-I Africa Consortium. Furthermore, they should highlight plans for the translation and delivery of the research findings to appropriate audiences through broad stakeholder engagement and include plans for ongoing and future community engagement activities and their potential benefits to relevant communities. Applicants should also describe how the Research Hub as a whole will facilitate research capacity building and career enhancement activities for researchers across the career spectrum. The applicant may wish to cross-reference other sections of the application where more detailed information is found (the Overall "Other Attachments" includes a diagram of the Research Hub Organizational Structure, for example). Other sections where interaction may be demonstrated include: the Administrative Core Research Strategy section, which details Hub management and cross-cutting activities; and the "Summary of Hub Data Management and Analysis" found in the Overall Resource Sharing section.

Letters of Support: Applications should include a letter of support from the appropriate institutional or organizational official (University or Medical School President, Dean or Director, or the head research administrator, company leader, government representative, or equivalent) from all collaborating organizations to substantiate the organizations' commitment to the proposed DS-I Africa Research Hub. Each organization should state its commitment to overcoming any administrative obstacles to the implementation of the application, such as accommodation for participation by multiple schools at a university or collaboration with other institutions within or outside of the applicant's country. Appropriate organizational commitment to the program includes the provision of adequate staff, facilities, and resources needed for the success of the planned program. Where applicable, the letter should address institutional commitment to any new faculty who participate in associated skill-building

programs or whose labs are established within the DS-I Africa Research Hub. It should also address commitment to DS-I Africa policies and procedures, including collaborating and sharing data with the DS-I Africa Consortium and Open Data Science Platform, as appropriate. The letter should briefly discuss the institution's plans for sustaining an active program of scientific research in the areas of this application following the end of the funding period. Where applicable, the letter should also address corporate interests and commitment to the proposed research project, applicable in-kind support, and goals for appropriate innovation commercialization.

The programmatic activities of this initiative will support national and international collaborations. Where relevant and attainable, letters of support from the related national ministries such as the Ministry of Science, Ministry of Health and/or Ministry of Education for each African country with a programmatic component are strongly encouraged. The letter should briefly describe the national policy concerning scientific research and health funding priorities.

Resource Sharing Plan: Individuals are required to comply with the instructions for the Resource Sharing Plans as provided in the SF424 (R&R) Application Guide, with the following modification:

A plan for data and resource sharing and release is expected for all NIH applications (https://grants.nih.gov/grants/policy/data_sharing/ (https://grants.nih.gov/grants/policy/data_sharing/)). Applicants are, therefore, expected to provide a well thought-out plan for sharing data and resources generated by the DS-I Africa Research Hub (including all program subcomponent projects and cores), which should later be aligned with the DS-I Africa Open Data Science Platform. Sharing of pre-existing data should be considered, where possible, for advancing consortium-level activities. Necessary exceptions to wide data sharing are allowable and should be discussed, including the need for controlled or restricted access for privacy, ethical, and intellectual property protections.

For the Overall Resource Sharing Plan, Hubs should summarize data-sharing activities and the sharing of other resources within and beyond the DS-I Africa Research Hub in a "Summary of Hub Data Management and Analysis".

Summary of Hub Data Management and Analysis: The applicant should provide a Summary of Hub Data Management and Analysis. This summary should include a general description of the types of data expected to be generated from the Hub and a synopsis of proposed data management/sharing activities. Applicants may wish to highlight anticipated integration opportunities between projects and cores. Information included in this section should briefly summarize, but not duplicate, data management plans from the component sections of the application; however, applicants are encouraged to cross-reference DMAC component and project/core Resource Sharing Plans for additional details about data management and analysis.

Appendix:

Only limited items are allowed in the Appendix. Follow all instructions for the Appendix as described in the SF424 (R&R) Application Guide; any instructions provided here are in addition to the SF424 (R&R) Application Guide instructions.

PHS Human Subjects and Clinical Trials Information (Overall)

When involving human subjects research, clinical research, and/or NIH-defined clinical trials follow all instructions for the PHS Human Subjects and Clinical Trials Information form in the SF424 (R&R) Application Guide, with the following additional instructions:

If you answered Yes to the question Are Human Subjects Involved? on the R&R Other Project Information form, there must be at least one human subjects study record using the **Study Record: PHS Human Subjects and Clinical Trials Information** form or a **Delayed Onset Study** record within the application. The study record(s) must be included in the component(s) where the work is being done, unless the same study spans multiple components. To avoid the creation of duplicate study records, a single study record with sufficient information for all involved components must be included in the Overall component when the same study spans multiple components.

Study Record: PHS Human Subjects and Clinical Trials Information

All instructions in the SF424 (R&R) Application Guide must be followed.

Delayed Onset Study

Note: [Delayed onset](https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy) (<https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy>) does NOT apply to a study that can be described but will not start immediately (i.e., delayed start).

All instructions in the SF424 (R&R) Application Guide must be followed.

PHS Assignment Request Form (Overall)

All instructions in the SF424 (R&R) Application Guide must be followed.

Administrative Core

When preparing your application, use Component Type Core.

All instructions in the SF424 (R&R) Application Guide must be followed, with the following additional instructions, as noted.

SF424 (R&R) Cover (Administrative Core)

Complete only the following fields:

- Applicant Information
- Type of Applicant (optional)
- Descriptive Title of Applicant's Project
- Proposed Project Start/Ending Dates

PHS 398 Cover Page Supplement (Administrative Core)

Enter Human Embryonic Stem Cells in each relevant component.

Research & Related Other Project Information (Administrative Core)

Human Subjects: Answer only the Are Human Subjects Involved? and 'Is the Project Exempt from Federal regulations?' questions.

Vertebrate Animals: Answer only the Are Vertebrate Animals Used? question.

Summary/Abstract: Include a Core Summary/Abstract that briefly describes how the Administrative Core will lead the Research Hub and ensure that the Hub's projects and cores are being supported to achieve the Hub's goals. The summary should also include specific activities and innovative plans for supporting research translation of the projects and cores within the Hubs. The summary should not be a general description of the requirements of the Administrative Core. Rather, applicants should highlight specific activities and innovative plans for meeting the core requirements, tailored to fit the overall goals of the Research Hub.

Project Narrative: Do not complete. Note: ASSIST screens will show an asterisk for this attachment indicating it is required. However, eRA systems only enforce this requirement in the Overall component and applications will not receive an error if omitted in other components.

Application guide states that Project Narrative is required. However, it is only required for the Overall component. If you would like the applicant to provide a project narrative for this component, update the above instructions accordingly. Specific names provided for Other Attachments must be no more than 50 characters including spaces.

Project /Performance Site Location(s) (Administrative Core)

List all performance sites that apply to the specific component.

Note: The Project Performance Site form allows up to 300 sites, prior to using additional attachment for additional entries.

Research & Related Senior/Key Person Profile (Administrative Core)

- In the Project Director/Principal Investigator section of the form, use Project Role of Other with Category of Core Lead and provide a valid eRA Commons ID in the Credential field.
- In the additional Senior/Key Profiles section, list Senior/Key persons that are working in the component.
- Include a single Biographical Sketch for each Senior/Key person listed in the application regardless of the number of components in which they participate. When a Senior/Key person is listed in multiple components, the Biographical Sketch can be included in any one component.
- If more than 100 Senior/Key persons are included in a component, the Additional Senior Key Person attachments should be used.
- The DS-I Africa Research Hub PD/PI (contact PD/PI for applications with multiple PDs/PIs) must serve as the Administrative Core Lead and the overall Hub Director, although additional leadership of other Projects or Cores is also permitted.
- Applicants may also designate an appropriate senior investigator as a Hub Deputy Director, with roles to be defined by the applicants.
- The Administrative Core is expected to have a qualified Hub Administrator to manage the day-to-day operations with responsibility for the administrative, budgetary, and operational aspects of the Hub. For the Hub Administrator, in the additional Senior/Key Profiles section, use Project Role of 'Other (Specify)' and provide the role under 'Other Project Role Category' as 'Hub Administrator'.

Budget (Administrative Core)

Budget forms appropriate for the specific component will be included in the application package.

Leadership Effort Commitment: The Research Hub Director must commit and maintain through the life of the award a minimum of 2.4 person-months of effort. The required levels of effort may reflect an aggregate of their effort across all Research Hub activities, as applicable, instead of solely effort pertaining to Administrative Core duties.

Hub Administrator: Applicants are expected to propose and budget for a full-time or part-time Hub Administrator to manage day-to-day operations, based on the complexity of the DS-I Africa Research Hub.

Travel Funds: The budget should include funds to support travel for Hub and Consortium activities, including but not limited to supporting the participation of PD/PI(s) and additional staff members at the biannual Consortium meetings. It is expected that each Research Hub will bring at least 2 people to each DS-I Africa Consortium meeting.

Other: Funds (including travel, if appropriate) may be allocated for expenses related to the formation of a Scientific Advisory Board.

Note: The R&R Budget form included in many of the component types allows for up to 100 Senior/Key Persons in section A and 100 Equipment Items in section C prior to using attachments for additional entries. All other SF424 (R&R) instructions apply.

PHS 398 Research Plan (Administrative Core)

Specific Aims: Specific Aims should describe the objectives and goals of the Core. The Specific Aims section should also contain a brief statement about how the Core will lead the Research Hub, how it will integrate a multisectoral Hub composed of diverse researchers and partners with biomedical, data science, engineering, etc. expertise; how it will lead the selection and management of pilot projects; how it will integrate the Hub with the broader DS-I Africa Consortium; how it will build and maintain vital partnerships; how it will regularly evaluate Hub activities to ensure quality and efficiency; and how it will conduct broad stakeholder engagement.

Research Strategy: The Administrative Core is expected to have appropriate and effective administrative and organizational capabilities to provide administrative support for multisectoral data science research and innovation, as well as for research opportunities and career enhancement for scientists across the career spectrum. It is also expected to foster synergy and integration of the DS-I Africa Research Hub components and partners and support any planning and evaluation activities that the applicant chooses to include. Applications must describe the structure of the Administrative Core utilizing the following subsections (instead of the standard sub-sections):

Sub-section A. Administration and Management: Provide an administrative plan appropriate for effective management of a complex, multisectoral Hub structure. Multiple PD/PI applications, if applicable, will additionally use the Multiple PD/PI Leadership Plan to further describe the rationale and justification for choosing the multiple PI approach, governance and organizational structure of the team, procedures for resolving conflicts, and process for making decisions on scientific direction and allocating resources and funds.

- Describe the governance and organizational structure of the leadership team, research projects, and all cores.
- Describe the plans for administering, managing, tracking, evaluating, and coordinating the activities of the Hub and its individual components.
- Delineate the roles and administrative, technical, and scientific responsibilities for the project/core staff as they relate to the various functions of the Hub.
- Include communication plans, processes for making decisions on scientific direction, and procedures for resolving conflicts.
- Include a plan for the integration of all Hub activities to accomplish overall goals.
- Describe operating procedures for ensuring responsiveness to members of the research community who wish to access the Hub and utilize Hub resources.
- Describe how the Administrative Core will oversee interaction between the Research Hub and other DS-I Africa Research Hubs, Training Programs, ELSI Projects, Open Data Science Platform, and Coordinating Center for advancing aligned research (as applicable), strengthened research capacity building efforts, informed community engagement, development of relevant ethical and legal frameworks, and open data sharing.
- Describe how the Administrative Core will provide appropriate ethical, legal, and social implications understanding and oversight to research projects and overall Hub activities and discuss potential issues that may arise.
- Describe how the Administrative Core will ensure appropriate and timely communication with the DS-I Africa Coordinating Center and Open Data Science Platform, the DS-I Africa Consortium (as applicable), and applicable NIH staff.

Sub-section B. Scientific Advisory Board: Each awarded DS-I Africa Research Hub must recruit, under the guidance of the Hub Director, external experts (i.e., from outside of the Hub) to serve as scientific advisors to the DS-I Africa Research Hub leadership. Applicants should describe the SAB's general composition, the range of expertise to be sought, and how the panel will be expected to contribute to the Hub's activities. However, to facilitate the review process, DO NOT name specific individuals in the application and DO NOT contact any potential candidates.

Sub-section C. Travel: A plan and budget for travel of Hub members to DS-I Africa Consortium meetings, internal Hub meetings, and other scientific meetings should be included.

Sub-section D. Pilot Projects: DS-I Africa Research Hubs are expected to select and evaluate promising data science-related pilot projects for sub-award funding that have significant potential to address developing needs and opportunities while being shorter term and varying throughout the period of award. A detailed plan for the process by which pilot projects will be solicited, reviewed, funded and managed must be provided.

The Pilot Projects Research Plan should provide the following information:

1. Plan for identifying and selecting new meritorious pilot projects.

- Informed by the proposed Research Hub's overarching focus and milestones, describe the process for soliciting and reviewing applications from within and outside of the Hub to be funded by the Hub. Hubs should describe a plan for soliciting applications from early-stage investigators as well as established investigators from diverse organizations (academia, industry, NGO, etc.).
- Describe the strategy for evaluating pilot project proposals and the process of prioritizing projects based on the Hub's resources and expertise. Hub leadership will establish review criteria and send the projects to reviewers with the expertise to review project applications.
- Describe the plan for selecting highly meritorious pilot projects that advance the strategic priorities of the Hub.
- *Hubs are required to fund two to five sub-award projects per year beginning in Year 2 of the research funding cycle*, with at least one sub-award project from outside the existing Hub organizational structure, in keeping with the overarching DS-I Africa goal of sustaining and expanding vital partnerships.
- The selection of sub-award projects for funding will be made in consultation with the Core leadership, additional Hub personnel, NIH programmatic and scientific staff, the Hub PD/PI(s) and applicable external scientific experts.

2. Develop a strategy for managing and assessing pilot projects.

- Briefly describe the Hub's strategy for overseeing funded pilot projects, including the distribution of yearly sub-award funds, especially for those that aren't located at the parent institution.
- Provide clear plans for monitoring and evaluating pilot project progress, in keeping with the overall goals of the DS-I Africa Research Hub.
- Describe strategy for aligning pilot project data sharing with other Research Hub activities for streamlined data management and analysis.
- Briefly describe potential plans for translating findings to diverse stakeholder groups as well as supporting further development of pilot projects through relevant partner organizations (as applicable).
- Describe plan for reporting pertinent information to appropriate NIH and DS-I Africa staff for informing potential use of common data elements to drive appropriate data harmonization and utilization across the DS-I Africa Consortium (as applicable).

Sub-section E. Research and Career Enhancement: Opportunities for involvement of scientists at all stages of the career pipeline, including students, post-docs, and early stage investigators, both from within and outside of the parent organization, should be described. Applicants should also consider what career enhancement opportunities will be provided, such as attending seminars and scientific meetings, writing papers, and giving talks. Long-term sustainability and institutional/governmental commitments to research education and independent career opportunities should also be discussed. Applicants should acknowledge that career development efforts performed at the Research Hub will be further aligned with the Consortium-wide efforts led by the DS-I Africa Research Training Program and should have a process in place to potentially mentor associated trainees (see [RFA-RM-20-016 \(https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-016.html\)](https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-016.html) for more information).

Letters of Support: Applicants may include letters of support pertaining to the Administrative Core. For example, these letters may include collaborative agreements, institutional support (if applicable), etc. See SF424 for details.

Resource Sharing Plan: Individuals are required to comply with the instructions for the Resource Sharing Plans as provided in the SF424 (R&R) Application Guide.

Appendix: Only limited items are allowed in the Appendix. Follow all instructions for the Appendix as described in the SF424 (R&R) Application Guide; any instructions provided here are in addition to the SF424 (R&R) Application Guide instructions.

PHS Human Subjects and Clinical Trials Information (Administrative Core)

When involving human subjects research, clinical research, and/or NIH-defined clinical trials follow all instructions for the PHS Human Subjects and Clinical Trials Information form in the SF424 (R&R) Application Guide, with the following additional instructions:

If you answered Yes to the question Are Human Subjects Involved? on the R&R Other Project Information form, you must include at least one human subjects study record using the **Study Record: PHS Human Subjects and Clinical Trials Information** form or a **Delayed Onset Study** record.

Study Record: PHS Human Subjects and Clinical Trials Information

All instructions in the SF424 (R&R) Application Guide must be followed.

Delayed Onset Study

Note: [Delayed onset \(https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy\)](https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy) does NOT apply to a study that can be described but will not start immediately (i.e., delayed start). All instructions in the SF424 (R&R) Application Guide must be followed.

Data Management and Analysis Core

When preparing your application in ASSIST, use Component Type 'Core.'

All instructions in the SF424 (R&R) Application Guide must be followed, with the following additional instructions, as noted.

SF424 (R&R) Cover (Data Management and Analysis Core)

Complete only the following fields:

- Applicant Information
- Type of Applicant (optional)
- Descriptive Title of Applicant's Project
- Proposed Project Start/Ending Dates

PHS 398 Cover Page Supplement (Data Management and Analysis Core)

Enter Human Embryonic Stem Cells in each relevant component.

Research & Related Other Project Information (Data Management and Analysis Core)

Human Subjects: Answer only the Are Human Subjects Involved? and 'Is the Project Exempt from Federal regulations?' questions.

Vertebrate Animals: Answer only the Are Vertebrate Animals Used? question.

Summary/Abstract: Include a Summary/Abstract that briefly describes how the DMAC will support the management and integration of data assets across the Hub.

Project Narrative: Do not complete. Note: ASSIST screens will show an asterisk for this attachment indicating it is required. However, eRA systems only enforce this requirement in the Overall component and applications will not receive an error if omitted in other components.

Project /Performance Site Location(s) (Data Management and Analysis Core)

List all performance sites that apply to the specific component.

Note: The Project Performance Site form allows up to 300 sites, prior to using additional attachment for additional entries.

Research & Related Senior/Key Person Profile (Data Management and Analysis Core)

- In the Project Director/Principal Investigator section of the form, use Project Role of Other with Category of Core Lead and provide a valid eRA Commons ID in the Credential field.
- In the additional Senior/Key Profiles section, list Senior/Key persons that are working in the component.

- Include a single Biographical Sketch for each Senior/Key person listed in the application regardless of the number of components in which they participate. When a Senior/Key person is listed in multiple components, the Biographical Sketch can be included in any one component.
- If more than 100 Senior/Key persons are included in a component, the Additional Senior Key Person attachments should be used

Budget (Data Management and Analysis Core)

Budget forms appropriate for the specific component will be included in the application package.

Note: The R&R Budget form included in many of the component types allows for up to 100 Senior/Key Persons in section A and 100 Equipment Items in section C prior to using attachments for additional entries. All other SF424 (R&R) instructions apply.

PHS 398 Research Plan (Data Management and Analysis Core)

Specific Aims: Specific Aims should describe the specific objectives and goals of the DMAC to support effective data management and analysis through development of a comprehensive data management plan including: 1) coordination with projects and cores; 2) fostering data sharing and interoperability; and 3) data quality assurance and quality control. The specific aims section should also contain a brief statement about how the Core integrates with each of the research projects and cores and helps to achieve the Hub's goals.

Research Strategy: The Research Strategy should include a Comprehensive Data Management Plan describing how the Core will serve as a resource for facilitating and supporting effective data management and analysis across the Research Hub for both use of existing data and the development of new data. For the purposes of this FOA, the DMAC would manage data generated from the Hub's research projects and cores as well as house previously compiled data sets, as applicable. The data would not be limited to "big data" (e.g., '-omics', geographic information data, or web-based data) but would also include smaller datasets. In addition, this section should include a description of the facilities, techniques, and professional skills/expertise provided by the core.

The Comprehensive Data Management Plan should include the following: 1) coordination with projects and cores; 2) fostering data sharing and interoperability; 3) data quality assurance and quality control. It is anticipated that this plan would evolve to incorporate new data and any changes in data management policies/practices.

1) Coordination with Projects and Cores. The DMAC should work closely with project/core leaders to assist with the development and refinement of individual project/core data management plans (see "Resource Sharing" section of project/core component). While it is highly encouraged that Research Hubs propose the leveraging of preexisting data sets for further analyses, DMACs should also include a description of the types of new data expected to be generated by the Hub's projects/cores, as applicable. The DMAC would work with project and core leaders in establishing a process for identifying which datasets hold a high priority for sharing versus data that may be more practical for internal curation (e.g., preliminary results from projects, etc) or that may require data sharing exceptions to protect intellectual property. The DMAC should assist project/core leaders in identifying appropriate data sharing platforms or strategies for sharing of prioritized data sets. This may include plans for how and in which form data will be shared (e.g. raw vs processed, de-identified) within and outside the Hub, and appropriate timelines for dataset deposition. It is anticipated that data science researchers may already have robust data management protocols that can be incorporated into their projects/cores. In these cases, the DMAC may incorporate project/core protocols as part of the DMAC's function. The DMAC should also indicate how it will support the data analytic, integration, and sharing efforts of each research project/core and how these interactions will occur over the course of the research projects/cores (e.g., monthly meetings with research project and core leaders). The DMAC may consider how it can encourage project/core leaders and trainees to analyze shared datasets, creating opportunities for exploring new linkages between data and the formation of new hypotheses. DMACs may also wish to develop a data management template (i.e., develop draft DMP for the project/core leads) to assist in organizing data assets with the projects/cores. The DMAC may indicate how it will coordinate with each research project/core leader on protection of intellectual property; assurance of and measures (steps taken) for data integrity and quality control; reporting of statistical analysis; timing for uploading data; and determining the most appropriate associated scientific discipline-endorsed vocabularies (i.e., ontologies). To facilitate these activities the core may wish to create a data management resource (e.g., a laboratory management information system) to capture these on-going interactions. Plans for training and education on data management and analysis practices may also be included. The DMAC should also designate a point of contact to coordinate with the DS-I Africa Open Data Science Platform staff and applicable NIH staff (in collaboration with the Hub's Administrative Core).

Additional Functionalities: In addition to the responsibilities outlined above, the DMAC may also include additional functionalities to assist projects/cores to advance the goals of data management and analysis. The DMAC may provide additional expertise for areas such as biostatistics, bioinformatics, geographical information systems, computational modeling, etc. These additional functionalities would be optional and should not detract from the data management and analysis responsibilities of the DMAC.

2) *Fostering Data Sharing and Interoperability.* The research strategy should address plans for sharing data and data resources within and outside the Research Hub, including which data repositories will be used. The DMAC should include discussion of ontology, data structures, standards, data platforms, interfaces, tools, and any data visualization approaches that may be used to facilitate data sharing and integration within the Hub. While it is not expected for the DMAC to create a repository for the Hub's data, the core should be familiar with databases that can be used to deposit the data (<https://fairsharing.org/> (<https://fairsharing.org/>)) and should develop a plan for project and core members to have access and use the data generated from the Hub, with appropriate ethical controls in place. The DMAC is highly encouraged to incorporate FAIR Principles when working with project/core leaders. Also, the DMAC should provide a plan for interfacing with any larger institutional information technology infrastructure(s) and addressing requirements for database structures and security. In addition, the DMAC may describe how metadata (e.g., data descriptors) and ontologies (or semantic relationships) will be used to harmonize data across the Hub to enable interoperability and compatibility and to identify the relationships between data. The DMAC should use common protocols where feasible and should capture experimental conditions (i.e., standardized data documentation) and analysis tools (e.g., codes and scripts). The DMAC may look for data coordination and sharing opportunities within the DS-I Africa Consortium and should plan to participate in applicable data management and analysis-focused working group calls.

Applicants should be aware of the DS-I Africa Open Data Science Platform (see companion FOA, [RFA-RM-20-018](https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-018.html) (<https://grants.nih.gov/grants/guide/rfa-files/RFA-RM-20-018.html>)), which will be a scalable data-sharing gateway available to the research community with access to disparate types of open and controlled-access data, generated from the DS-I Africa Research Hubs as well as other existing sources. Besides democratizing access to and use of data through the use of FAIR Principles, the Open Data Science Platform will provide connectivity among the DS-I Africa participants, serve as a technical resource for the Consortium, and provide a core set of tools that enable cross-consortium projects and have the ability to deploy computational pipelines, workflows, and analyses developed by the Research Hub investigators and other users to analyze their own data in conjunction with other data accessible via the Open Data Science Platform. All DS-I Africa-funded projects are required to and will benefit from interaction with the DS-I Africa Open Data Science Platform. Therefore, applicants for a research project funded under this FOA must include a statement from the applicant institution committing it to collaborating and, where possible, sharing data access (open or controlled) with the DS-I Africa Open Data Science Platform, consistent with achieving the goals of DS-I Africa.

3) *Data Quality Assurance and Quality Control:* The Research Strategy section should include a "Data Quality Assurance and Quality Control" statement that is applicable to the data types generated by the Hub. This section should include the Hub's policy on implementing and assessing the effectiveness of its operations (for an example, see DataONE, <https://www.dataone.org/best-practices/develop-quality-assurance-and-quality-control-plan> (<https://www.dataone.org/best-practices/develop-quality-assurance-and-quality-control-plan>)) and include:

- How the core's activities will be reviewed and evaluated both internally and externally to ensure quality, including any plans for peer or other reviews of the study design or analytical methods prior to data collection and how recommended changes will be implemented.
- A brief description of data quality assurance and control procedures.
- How a project/core staff and trainees will be trained to implement quality assurance, and who will be responsible for training.

If the Research Hub only proposes the use of pre-existing data with no collection of new data, this section should clearly state how these data will be managed appropriately and stored securely, as well as how assessments utilizing these data will be performed and validated.

In the case that proposed research activities require research infrastructure and/or tools (such as data platforms, data storage, high bandwidth access, cloud computing, etc.) beyond what is currently available through the Hub and its collaborating institutions, the DMAC should list, provide appropriate justifications for expenditures associated with infrastructure investments, and discuss how and when any infrastructure development would take place to limit delays in proposed research.

Relation to Overall Research Hub: The DMAC should also include a "Relation to Overall Research Hub" statement. In this statement, clearly state the relevance of the core to the goals of the overall Hub; how this core integrates with other projects and cores; and how the findings/activities of this core assist in solving the problem(s) that the Hub is addressing. As applicable, applicants may cross-reference the "Summary of Research Hub Data Management and Analysis" (see Resource Sharing Plan of the Overall Component) as well as the Data Management Plan for each project and core (see Resource Sharing Plan for Projects and Cores).

Letters of Support: Applicants may include letters to demonstrate the support of, for example, research collaborators, stakeholders, potential end-users, and/or entities whose resources may be utilized as part of the Core (e.g. data set sharing, shared research infrastructure, etc.). It is recommended that applicants include letters from any applicable data repository

facilities and institutional support (if applicable) when leveraging such resources for the DMAC. See SF424 (R&R) Application Guide for details.

Resource Sharing Plan: Individuals are required to comply with the instructions for the Resource Sharing Plans as provided in the SF424 (R&R) Application Guide.

Appendix: Only limited items are allowed in the Appendix. Follow all instructions for the Appendix as described in the SF424 (R&R) Application Guide; any instructions provided here are in addition to the SF424 (R&R) Application Guide instructions.

PHS Human Subjects and Clinical Trials Information (Data Management and Analysis Core)

When involving human subjects research, clinical research, and/or NIH-defined clinical trials follow all instructions for the PHS Human Subjects and Clinical Trials Information form in the SF424 (R&R) Application Guide, with the following additional instructions:

If you answered Yes to the question Are Human Subjects Involved? on the R&R Other Project Information form, you must include at least one human subjects study record using the **Study Record: PHS Human Subjects and Clinical Trials Information** form or a **Delayed Onset Study** record.

Study Record: PHS Human Subjects and Clinical Trials Information

All instructions in the SF424 (R&R) Application Guide must be followed

Delayed Onset Study

Note: [Delayed onset \(https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy\)](https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy) does NOT apply to a study that can be described but will not start immediately (i.e., delayed start). All instructions in the SF424 (R&R) Application Guide must be followed.

Support Core (Optional)

When preparing your application in ASSIST, use Component Type 'Core.'

All instructions in the SF424 (R&R) Application Guide must be followed, with the following additional instructions, as noted.

SF424 (R&R) Cover (Support Core)

Complete only the following fields:

- Applicant Information
- Type of Applicant (optional)
- Descriptive Title of Applicant's Project
- Proposed Project Start/Ending Dates

PHS 398 Cover Page Supplement (Support Core)

Enter Human Embryonic Stem Cells in each relevant component.

Research & Related Other Project Information (Support Core)

Human Subjects: Answer only the Are Human Subjects Involved? and 'Is the Project Exempt from Federal regulations?' questions.

Vertebrate Animals: Answer only the Are Vertebrate Animals Used? question.

Summary/Abstract: Include a Core Summary/Abstract that briefly describes how the optional Support Core will support the Research Hub. The summary should highlight specific activities, innovative plans, and unique capabilities of the core indicating how its objectives have been tailored to fit the overall goals of the Hub.

Project Narrative: Do not complete. Note: ASSIST screens will show an asterisk for this attachment indicating it is required. However, eRA systems only enforce this requirement in the Overall component and applications will not receive an error if omitted in other components.

Project /Performance Site Location(s) (Support Core)

List all performance sites that apply to the specific component.

Note: The Project Performance Site form allows up to 300 sites, prior to using additional attachment for additional entries.

Research & Related Senior/Key Person Profile (Support Core)

- In the Project Director/Principal Investigator section of the form, use Project Role of Other with Category of Core Lead and provide a valid eRA Commons ID in the Credential field.
- In the additional Senior/Key Profiles section, list Senior/Key persons that are working in the component.
- Include a single Biographical Sketch for each Senior/Key person listed in the application regardless of the number of components in which they participate. When a Senior/Key person is listed in multiple components, the Biographical Sketch can be included in any one component.
- If more than 100 Senior/Key persons are included in a component, the Additional Senior Key Person attachments should be used

Budget (Support Core)

Budget forms appropriate for the specific component will be included in the application package.

Note: The R&R Budget form included in many of the component types allows for up to 100 Senior/Key Persons in section A and 100 Equipment Items in section C prior to using attachments for additional entries. All other SF424 (R&R) instructions apply.

PHS 398 Research Plan (Support Core)

Specific Aims: Specific Aims should describe the objectives and goals of the Core. The specific aims section should also contain a brief statement about how the Core integrates with the rest of the Research Hub.

Research Strategy: In the Research Strategy, the applicant must include the description of the optional Support Core, the services to be rendered, the methodological approaches to be used, and a plan for prioritizing the use of the Core by Hub members. This section must clearly present the facilities, techniques, and professional skills that the core will provide. Explain how the Core is justified by the scientific and translational needs of the DS-I Africa Research Hub. Describe how the expected contributions of the Core will accelerate progress and promote more effective approaches that will enhance the likelihood of success of the research project(s) in achieving the overall objectives of the DS-I Africa Research Hub. Explain why support for the indicated functionality in the form of a centralized Core of the DS-I Africa Research Hub will be more efficient than if the research projects themselves were to provide the functionality or how the Core provides added value beyond that which would be provided by acquisition through fee for service or commercial access. For a Core that by its nature is not innovative, describe why it is essential to achieve the goals or objectives of the DS-I Africa Research Hub.

Describe plans and procedures for establishing and managing the Core. Describe the role of the Core Leader and each of the key participants. Describe plans and procedures for quality control and cost-effectiveness of the services and resources provided by the Core to the DS-I Africa Research Hub investigators. If the Support Core is designed to interface with the sub-award pilot projects, describe the process(es) that will be in place to facilitate coordination, distribution and accessibility to the services, facilities or resources that the Core proposes to provide.

In the Research Strategy section, a Support Core that provides analytical and quantitative services to the applicant's Hub should include a Quality Assurance Section. The Quality Assurance Section should document the Hub's policy on implementing and assessing the effectiveness of its quality assurance and quality control operations.

The Quality Assurance Section should include the following items such as:

- How the core's activities will be reviewed and evaluated both internally and externally to ensure quality, including any plans for peer or other reviews of the study design or analytical methods prior to data collection and how recommended changes will be implemented.
- How a project's staff will be trained to implement quality assurance, and who will be responsible for training.
- How data will be analyzed (e.g., data/statistical analysis methods and references to software) and plans for coordinating with the DMAC.
- Discussion of any computer models to be designed or utilized with associated verification and validation techniques.
- How data will be managed (collected, backed-up, collated, transferred, stored, documented, and shared as appropriate) and coordinated with the DMAC.

Relation to Overall Research Hub: The optional Support Core should also include a "Relation to Overall Research Hub" statement. In this statement, clearly state the relevance of the core to the goals of the overall Hub, how this core integrates with other projects and cores, and how the findings/activities of this core assist in solving the problem(s) that the Hub is addressing. Interactions with the Data Management and Analysis Core should also be included.

Resource Sharing Plan: Individuals are required to comply with the instructions for the Resource Sharing Plans as provided in the SF424 (R&R) Application Guide.

Appendix: Only limited items are allowed in the Appendix. Follow all instructions for the Appendix as described in the SF424 (R&R) Application Guide; any instructions provided here are in addition to the SF424 (R&R) Application Guide instructions.

PHS Human Subjects and Clinical Trials Information (Support Core)

When involving human subjects research, clinical research, and/or NIH-defined clinical trials follow all instructions for the PHS Human Subjects and Clinical Trials Information form in the SF424 (R&R) Application Guide, with the following additional instructions:

If you answered Yes to the question Are Human Subjects Involved? on the R&R Other Project Information form, you must include at least one human subjects study record using the **Study Record: PHS Human Subjects and Clinical Trials Information** form or a **Delayed Onset Study** record.

Study Record: PHS Human Subjects and Clinical Trials Information

All instructions in the SF424 (R&R) Application Guide must be followed

Delayed Onset Study

Note: [Delayed onset \(https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy\)](https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy) does NOT apply to a study that can be described but will not start immediately (i.e., delayed start). All instructions in the SF424 (R&R) Application Guide must be followed.

Research Project

A DS-I Africa Research Hub must comprise *at least two connected research projects* in Africa that together address one or more research areas of interest to the applicant. Each Project should be described as a distinct component.

When preparing your application, use Component Type Project.

All instructions in the SF424 (R&R) Application Guide must be followed, with the following additional instructions, as noted.

SF424 (R&R) Cover (Research Project)

Complete only the following fields:

- Applicant Information
- Type of Applicant (optional)
- Descriptive Title of Applicant's Project
- Proposed Project Start/Ending Dates

PHS 398 Cover Page Supplement (Research Project)

Enter Human Embryonic Stem Cells in each relevant component.

Research & Related Other Project Information (Research Project)

Human Subjects: Answer only the Are Human Subjects Involved? and 'Is the Project Exempt from Federal regulations? questions.

Vertebrate Animals: Answer only the Are Vertebrate Animals Used? question.

Summary/Abstract: Include a Project Summary/Abstract that briefly describes the objectives of the project and how the project will support the data science and health innovation goals of the Research Hub. Relate the project to the DS-I Africa program's overarching goals.

Provide details about the project such as the health problem being evaluated, research approaches, applicable data sets that can be leveraged, etc. In addition, the project summary should highlight aspects of the application that are innovative, paradigm-shifting, and/or gap-filling. A brief summary of plans for engaging stakeholders or communities, if applicable, should be included. The project summary should provide a depth of understanding of the science and activities of the project written in plain language for a diverse set of reviewers.

Project Narrative: Do not complete. Note: ASSIST screens will show an asterisk for this attachment indicating it is required. However, eRA systems only enforce this requirement in the Overall component and applications will not receive an error if omitted in other components.

Project /Performance Site Location(s) (Research Project)

List all performance sites that apply to the specific component.

Note: The Project Performance Site form allows up to 300 sites, prior to using additional attachment for additional entries.

Research & Related Senior/Key Person Profile (Research Project)

- In the Project Director/Principal Investigator section of the form, use Project Role of Other with Category of Project Lead and provide a valid eRA Commons ID in the Credential field.
- In the additional Senior/Key Profiles section, list Senior/Key persons that are working in the component.
- Include a single Biographical Sketch for each Senior/Key person listed in the application regardless of the number of components in which they participate. When a Senior/Key person is listed in multiple components, the Biographical Sketch can be included in any one component.
- If more than 100 Senior/Key persons are included in a component, the Additional Senior Key Person attachments should be used.
- The Project Lead investigators' experiences with conducting research and employing the proposed methods and technologies should be documented.

Budget (Research Project)

Budget forms appropriate for the specific component will be included in the application package.

Note: The R&R Budget form included in many of the component types allows for up to 100 Senior/Key Persons in section A and 100 Equipment Items in section C prior to using attachments for additional entries. All other SF424 (R&R) instructions apply.

PHS 398 Research Plan (Research Project)

Specific Aims: Specific Aims should describe the objectives and goals of the project. The specific aims section should also contain a brief statement about how the project relates with the overall goals of the proposed Research Hub and integrates with the rest of the Research Hub, describing specific aims shared between projects and cores, as applicable.

Research Strategy: The Research Strategy section should highlight the significance of the research project, the expertise of the research team, the innovation of the science, the details of the experimental approach (including a rationale for the methodology), the strengths of the research environment, as well as a description of how the project contributes to the Overall Research Hub. As such, the following subsections are recommended: Significance and Relevance to the DS-I Africa Program; Investigators; Innovation; Approach; Environment; and Research Hub Integration.

The significance of the project should be described, including the specific contribution that the project will make to the overall success of the DS-I Africa Research Hub. The research strategy that will be pursued in the project should be described in this section. While new data may be collected, it cannot be the main focus of the research project and projects should incorporate re-use of data as much as possible. As such, Hubs should leverage existing, African-based information and communication technology infrastructure, such as cell coverage and mobile phones, satellites, high-performance computing infrastructure and other digital information systems whenever possible. If new data are to be generated by the research project, the costs of generating the data should be addressed, as should the availability of supplies and any issues of long-term maintenance and servicing of necessary equipment. Research Hubs should demonstrate through these projects the long-term potential for data science-driven research products to impact health, healthcare, or health research and translate beyond the research lab.

The experimental plan that will be employed should be discussed in detail, including methods and technologies. The data to be analyzed should be identified and the research strategy must include a plan for data acquisition (where applicable), storage, etc. Whether the data are new, previously generated, or publicly available, and whether new data are produced in-house or obtained externally, the applicant should describe how they will be analyzed, as well as the research team's experience with the relevant data science methodologies. If technology/solution development is proposed, the applicant should describe how it will be assessed and validated, as well as the research team's experience developing related technology platforms and implementing associated solutions or interventions. Applicants should align efforts with other projects and cores, specifically the Data Management and Analysis Core (DMAC), and state their willingness to work with the DS-I Africa Open Data Science Platform and other grantees for data harmonization, where appropriate, for the specific question being studied. Ideally this will occur early in the progression of the award, so pertinent new data are able to be harmonized from the beginning.

The data science and innovation needs of the project should be clearly described, including issues of data acquisition, data management, data storage, analytical capability, technology development, intervention validation, etc.; the data science- and/or innovation-related tools that will be used or developed, and the relevant investigators' multidisciplinary experience. The data science and innovation activities may be embedded within the individual research components but should at least be aligned with the DMAC activities, if not led collaboratively with researchers within the DMAC.

The applicant should identify potential major challenges or problems that might be encountered in conducting the proposed research and discuss how these would be addressed should they arise. Innovative approaches that will be developed or employed by the project should be identified and the impact of those methods/technologies on the future of African science, public health and/or medicine should be briefly discussed.

Research project milestones must be included for each project, defining how the applicants propose to achieve the key goals of the project. An associated timeline should provide further information regarding the timeframe of each element associated with each project milestone. These milestones and associated timeline should be distinct from the overall Research Hub milestones, although aligned, and more granular to address how applicants propose to achieve the specific goals of each research project.

Relation to Overall Research Hub: Applicants should include a description that clearly states the relevance of the project to the goals of the overall Research Hub, how this project integrates with other projects and cores, and how the findings/activities of this project assist in solving the problem(s) that the Hub is addressing. Coordination and or shared specific aims should be cross-referenced between projects/cores. Interactions with the Administrative Core's pilot projects and Data Management and Analysis Core should also be included.

Letters of Support: While not repeating letters of support associated with the Overall Research Hub, applicants may include letters to demonstrate the support of, for example, research collaborators, stakeholders, and/or entities whose resources may be utilized as part of the Project (e.g. data set sharing, shared research infrastructure, etc.). Letters from appropriate stakeholders vital to the success of the proposed research projects or those highlighting the unique capabilities of the collaborative research team and environment for addressing the associated research goals are also encouraged.

Resource Sharing Plan: Individuals are required to comply with the instructions for the Resource Sharing Plans as provided in the SF424 (R&R) Application Guide. Projects are expected to include a Data Management Plan within their Data Sharing Plan.

The "Data Management Plan" should describe the plans for data management and should be cross-referenced in the Data Management and Analysis Core.

All applications, regardless of the amount of direct costs requested for any one year, should include a Data Sharing Plan. The application must include a statement that the investigators and applicant institutions will abide by the Consortium's data sharing policies, which will be mutually agreed upon during the first year of the DS-I Africa Consortium's activities.

Appendix: Only limited items are allowed in the Appendix. Follow all instructions for the Appendix as described in the SF424 (R&R) Application Guide.

PHS Human Subjects and Clinical Trials Information (Research Project)

When involving human subjects research, clinical research, and/or NIH-defined clinical trials follow all instructions for the PHS Human Subjects and Clinical Trials Information form in the SF424 (R&R) Application Guide, with the following additional instructions:

If you answered Yes to the question Are Human Subjects Involved? on the R&R Other Project Information form, you must include at least one human subjects study record using the **Study Record: PHS Human Subjects and Clinical Trials Information** form or a **Delayed Onset Study** record.

Study Record: PHS Human Subjects and Clinical Trials Information

All instructions in the SF424 (R&R) Application Guide must be followed.

Delayed Onset Study

Note: [Delayed onset \(https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy\)](https://grants.nih.gov/grants/glossary.htm#DelayedOnsetHumanSubjectStudy) does NOT apply to a study that can be described but will not start immediately (i.e., delayed start). All instructions in the SF424 (R&R) Application Guide must be followed.

Foreign Institutions

Foreign (non-U.S.) institutions must follow policies described in the [NIH Grants Policy Statement \(https://grants.nih.gov/grants/guide/uri_redirect.htm?id=11137\)](https://grants.nih.gov/grants/guide/uri_redirect.htm?id=11137), and procedures for foreign institutions described throughout the SF424 (R&R) Application Guide.

3. Unique Entity Identifier and System for Award Management (SAM)

See Part 1. Section III.1 for information regarding the requirement for obtaining a unique entity identifier and for completing and maintaining active registrations in System for Award Management (SAM), NATO Commercial and Government Entity (NCAGE) Code (if

applicable), eRA Commons, and Grants.gov.

4. Submission Dates and Times

[Part I. Overview Information](#) contains information about Key Dates and times. Applicants are encouraged to submit applications before the due date to ensure they have time to make any application corrections that might be necessary for successful submission. When a submission date falls on a weekend or [Federal holiday \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=82380\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=82380), the application deadline is automatically extended to the next business day.

Organizations must submit applications to [Grants.gov \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11128\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11128) (the online portal to find and apply for grants across all Federal agencies) using ASSIST or other electronic submission systems. Applicants must then complete the submission process by tracking the status of the application in the [eRA Commons \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11123\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11123), NIH's electronic system for grants administration. NIH and Grants.gov systems check the application against many of the application instructions upon submission. Errors must be corrected and a changed/corrected application must be submitted to Grants.gov on or before the application due date and time. If a Changed/Corrected application is submitted after the deadline, the application will be considered late. Applications that miss the due date and time are subjected to the NIH Policy on Late Application Submission.

Applicants are responsible for viewing their application before the due date in the eRA Commons to ensure accurate and successful submission.

Information on the submission process and a definition of on-time submission are provided in the SF424 (R&R) Application Guide.

5. Intergovernmental Review (E.O. 12372)

This initiative is not subject to [intergovernmental review \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11142\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11142).

6. Funding Restrictions

All NIH awards are subject to the terms and conditions, cost principles, and other considerations described in the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11120\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11120).

Pre-award costs are allowable only as described in the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11143\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11143).

7. Other Submission Requirements and Information

Applications must be submitted electronically following the instructions described in the SF424 (R&R) Application Guide. Paper applications will not be accepted.

For information on how your application will be automatically assembled for review and funding consideration after submission go to: http://grants.nih.gov/grants/ElectronicReceipt/files/Electronic_Multi-project_Application_Image_Assembly.pdf ([//grants.nih.gov/grants/ElectronicReceipt/files/Electronic_Multi-project_Application_Image_Assembly.pdf](https://grants.nih.gov/grants/ElectronicReceipt/files/Electronic_Multi-project_Application_Image_Assembly.pdf)).

Applicants must complete all required registrations before the application due date. [Section III. Eligibility Information](#) contains information about registration.

For assistance with your electronic application or for more information on the electronic submission process, visit [How to Apply Application Guide \(https://grants.nih.gov/grants/how-to-apply-application-guide.html\)](https://grants.nih.gov/grants/how-to-apply-application-guide.html). If you encounter a system issue beyond your control that threatens your ability to complete the submission process on-time, you must follow the [Dealing with System Issues \(https://grants.nih.gov/grants/how-to-apply-application-guide/due-dates-and-submission-policies/dealing-with-system-issues.htm\)](https://grants.nih.gov/grants/how-to-apply-application-guide/due-dates-and-submission-policies/dealing-with-system-issues.htm) guidance. For assistance with application submission, contact the Application Submission Contacts in [Section VII](#).

Important reminders:

All PD(s)/PI(s) and component Project Leads must include their eRA Commons ID in the Credential field of the Senior/Key Person Profile Component of the SF424(R&R) Application Package. Failure to register in the Commons and to include a valid PD/PI Commons ID in the credential field will prevent the successful submission of an electronic application to NIH.

The applicant organization must ensure that the DUNS number it provides on the application is the same number used in the organization's profile in the eRA Commons and for the System for Award Management (SAM). Additional information may be found in the SF424 (R&R) Application Guide.

See [more tips \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11146\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11146) for avoiding common errors.

Upon receipt, applications will be evaluated for completeness and compliance with application instructions by the Center for Scientific Review and responsiveness by [components of participating organizations](#), NIH. Applications that are incomplete, non-compliant and/or nonresponsive will not be reviewed.

Use of Common Data Elements in NIH-funded Research

Many NIH ICs encourage the use of common data elements (CDEs) in basic, clinical, and applied research, patient registries, and other human subject research to facilitate broader and more effective use of data and advance research across studies. CDEs are data elements that have been identified and defined for use in multiple data sets across different studies. Use of CDEs can facilitate data sharing and standardization to improve data quality and enable data integration from multiple studies and sources, including electronic health records. NIH ICs have identified CDEs for many clinical domains (e.g., neurological disease), types of studies (e.g. genome-wide association studies (GWAS)), types of outcomes (e.g., patient-reported outcomes), and patient registries (e.g., the Global Rare Diseases Patient Registry and Data Repository). NIH has established a Common Data Element (CDE) Resource Portal" (<http://cde.nih.gov/> (<http://cde.nih.gov/>)) to assist investigators in identifying NIH-supported CDEs when developing protocols, case report forms, and other instruments for data collection. The Portal provides guidance about and access to NIH-supported CDE initiatives and other tools and resources for the appropriate use of CDEs and data standards in NIH-funded research. Investigators are encouraged to consult the Portal and describe in their applications any use they will make of NIH-supported CDEs in their projects.

Post Submission Materials

Applicants are required to follow the instructions for post-submission materials, as described in [the policy](#) (http://grants.nih.gov/grants/guide/url_redirect.htm?id=82299). Any instructions provided here are in addition to the instructions in the policy.

Section V. Application Review Information

1. Criteria

Only the review criteria described below will be considered in the review process. Applications submitted to the NIH in support of the [NIH mission](#) (http://grants.nih.gov/grants/guide/url_redirect.htm?id=11149) are evaluated for scientific and technical merit through the NIH peer review system.

For this announcement, each application will be evaluated both on the basis of the overall impact of the application as a whole and the overall impact of each of the individual components. Projects will receive individual impact scores and will also receive criterion scores on significance, investigators, innovation, approach, and environment. Cores will be evaluated based on the particular review criteria outlined in its section. Finally, the overall application will receive an overall impact score reflecting the reviewers' assessment of the Hub's Significance, Investigator(s), Innovation, Approach, Environment, and Integration as well as the likelihood the DS-I Africa Research Hub will be successful in accomplishing its goals.

In addition, for applications involving clinical trials: A proposed Clinical Trial application may include study design, methods, and intervention that are not by themselves innovative but address important questions or unmet needs. Additionally, the results of the clinical trial may indicate that further clinical development of the intervention is unwarranted or lead to new avenues of scientific investigation.

Overall Impact - Overall

Reviewers will provide an overall impact score to reflect their assessment of the likelihood for the DS-I Africa Research Hub to exert a sustained, powerful influence on the research field(s) involved, in consideration of the following review criteria and additional review criteria (as applicable for the DS-I Africa Research Hub proposed).

Scored Review Criteria - Overall

Reviewers will consider each of the review criteria below in the determination of scientific merit, and give a separate score for each. An application does not need to be strong in all categories to be judged likely to have major scientific impact. For example, a DS-I Africa Research Hub that by its nature is not innovative may be essential to advance a field.

Significance

Does the DS-I Africa Research Hub address an important problem or a critical barrier to progress in the field? Is the prior research that serves as the key support for the proposed project rigorous? If the aims of the DS-I Africa Research Hub are achieved, how will scientific knowledge, technical capability, and/or clinical practice be improved? How will successful completion of the aims change the concepts, methods, technologies, treatments, services, or preventative interventions that drive this field?

Specific for this FOA: Does the proposed Research Hub adequately address the objectives of the DS-I Africa program more broadly? If the Hub were successful, would it lead to an incremental advance, or would it provide a substantial step forward?

In addition, for applications involving clinical trials:

Are the scientific rationale and need for a clinical trial to test the proposed hypothesis or intervention well supported by preliminary data, clinical and/or preclinical studies, or information in the literature or knowledge of biological mechanisms? For trials focusing on clinical or public health endpoints, is this clinical trial necessary for testing the safety, efficacy or effectiveness of an intervention that could lead to a change in clinical practice, community behaviors or health care policy? For trials focusing on mechanistic, behavioral, physiological, biochemical, or other biomedical endpoints, is this trial needed to advance scientific understanding?

Investigator(s)

Are the PD(s)/PI(s), collaborators, and other researchers well suited to the DS-I Africa Research Hub? If Early Stage Investigators or those in the early stages of independent careers, do they have appropriate experience and training? If established, have they demonstrated an ongoing record of accomplishments that have advanced their field(s)? If the project is collaborative or multi-PD/PI, do the investigators have complementary and integrated expertise; are their leadership approach, governance and organizational structure appropriate for the project?

Specific for this FOA: Are the proposed collaborators likely to work together synergistically and enhance the potential impact of the research and activities proposed? Do the experience and scientific leadership of the Research Hub's Director(s) (PD/PI(s)) allow him/her to effectively direct a large complex multicomponent Hub and coordinate the interactions of the research projects with effective utilization of Cores to achieve programmatic goals? Is there evidence the Hub Director(s) has brought together complementary and integrated expertise, among research project and core leaders, to accomplish the goals of the Hub?

In addition, for applications involving clinical trials:

With regard to the proposed leadership for the project, do the PD/PI(s) and key personnel have the expertise, experience, and ability to organize, manage and implement the proposed clinical trial and meet milestones and timelines? Do they have appropriate expertise in study coordination, data management and statistics? For a multicenter trial, is the organizational structure appropriate and does the application identify a core of potential center investigators and staffing for a coordinating center?

Innovation

Does the application challenge and seek to shift current research or clinical practice paradigms by utilizing novel theoretical concepts, approaches or methodologies, instrumentation, or interventions? Are the concepts, approaches or methodologies, instrumentation, or interventions novel to one field of research or novel in a broad sense? Is a refinement, improvement, or new application of theoretical concepts, approaches or methodologies, instrumentation, or interventions proposed?

Specific for this FOA: Does the proposed Research Hub sufficiently promote and support innovative approaches or technologies that will strengthen and sustain data science efforts in Africa? Does the Hub propose to utilize novel theoretical concepts, approaches or methodologies, instrumentation, or interventions to solve complex problems associated with data science for health discovery and innovation in Africa?

In addition, for applications involving clinical trials:

Does the design/research plan include innovative elements, as appropriate, that enhance its sensitivity, potential for information or potential to advance scientific knowledge or clinical practice?

Approach

Are the overall strategy, methodology, and analyses well-reasoned and appropriate to accomplish the specific aims of the DS-I Africa Research Hub? Have investigators included plans to address weaknesses in the rigor of prior research that serves as the key support for the proposed project? Have the investigators presented strategies to ensure a robust and unbiased approach, as appropriate for the work proposed? Are potential problems, alternative strategies, and benchmarks for success presented? If the project is in the early stages of development, will the strategy establish feasibility and will particularly risky aspects be managed? Have the investigators presented adequate plans to address relevant biological variables, such as sex, for studies in vertebrate animals or human subjects?

If the DS-I Africa Research Hub involves human subjects and/or NIH-defined clinical research, are the plans to address:

- 1) the protection of human subjects from research risks, and
- 2) inclusion (or exclusion) of individuals on the basis of sex/gender, race, and ethnicity, as well as the inclusion or exclusion of individuals of all ages (including children and older adults), justified in terms of the scientific goals and research strategy proposed?

Specific for this FOA: Will the program proposed be effective for enhancing the competitiveness of African data science investigators? Will the Research Hub support the capacity building objectives of DS-I Africa? Are the evaluation plans, milestones and timelines proposed appropriate and adequate for the program? Has the issue of future sustainability been adequately addressed? Does the Research Hub describe appropriate strategies for community engagement with clear objectives of what they expect to achieve? Are there adequate plans to evaluate whether the strategies to engage the community are successful and the objectives achieved? Are plans for maintaining and engaging partners and collaborators within the Research Hub adequate to support the planned aims of the Hub?

In addition, for applications involving clinical trials:

Does the application adequately address the following, if applicable:

Study Design

Is the study design justified and appropriate to address primary and secondary outcome variable(s)/endpoints that will be clear, informative and relevant to the hypothesis being tested? Is the scientific rationale/premise of the study based on previously well-designed preclinical and/or clinical research? Given the methods used to assign participants and deliver interventions, is the study design adequately powered to answer the research question(s), test the proposed hypothesis/hypotheses, and provide interpretable results? Is the trial appropriately designed to conduct the research efficiently? Are the study populations (size, gender, age, demographic group), proposed intervention arms/dose, and duration of the trial, appropriate and well justified?

Are potential ethical issues adequately addressed? Is the process for obtaining informed consent or assent appropriate? Is the eligible population available? Are the plans for recruitment outreach, enrollment, retention, handling dropouts, missed visits, and losses to follow-up appropriate to ensure robust data collection? Are the planned recruitment timelines feasible and is the plan to monitor accrual adequate? Has the need for randomization (or not), masking (if appropriate), controls, and inclusion/exclusion criteria been addressed? Are differences addressed, if applicable, in the intervention effect due to sex/gender and race/ethnicity?

Are the plans to standardize, assure quality of, and monitor adherence to, the trial protocol and data collection or distribution guidelines appropriate? Is there a plan to obtain required study agent(s)? Does the application propose to use existing available resources, as applicable?

Data Management and Statistical Analysis

Are planned analyses and statistical approach appropriate for the proposed study design and methods used to assign participants and deliver interventions? Are the procedures for data management and quality control of data adequate at clinical site(s) or at center laboratories, as applicable? Have the methods for standardization of procedures for data management to assess the effect of the intervention and quality control been addressed? Is there a plan to complete data analysis within the proposed period of the award?

Environment

Will the scientific environment in which the work will be done contribute to the probability of success? Are the institutional support, equipment and other physical resources available to the investigators adequate for the project proposed? Will the project benefit from unique features of the scientific environment, subject populations, or collaborative arrangements?

Specific for this FOA: Does the Research Hub take advantage of resources available at the collaborating institutions? Does the application identify and plan to take advantage of other resources at the participants' institutions, countries or other countries where necessary? Do the letters of collaboration and institutional support show strong commitment to the project and to the PD(s)/PI(s)? Is there potential to enhance the sustainability of the proposed research and research products through the ecosystem created by the Hub partnership structure? Is there a commitment at the institution to providing quality research education and career development for early stage scientists?

In addition, for applications involving clinical trials:

If proposed, are the administrative, data coordinating, enrollment and laboratory/testing centers, appropriate for the trial proposed? Does the application adequately address the capability and ability to conduct the trial at the proposed site(s) or centers? Are the plans to add or drop enrollment centers, as needed, appropriate? If international site(s) is/are proposed, does the application adequately address the complexity of executing the clinical trial? If multi-sites/centers, is there evidence of the ability of the individual site or center to: (1) enroll the proposed numbers; (2) adhere to the protocol; (3) collect and transmit data in an accurate and timely fashion; and, (4) operate within the proposed organizational structure?

Additional Review Criteria - Overall

As applicable for the DS-I Africa Research Hub proposed, reviewers will evaluate the following additional items while determining scientific and technical merit, and in providing an overall impact score, but will not give separate scores for these items.

Integration of the Research Hub

Is the proposed collaborative nature of the Research Hub conducive to achieve the Hub's goals? Does the application provide adequate evidence of the advantages of conducting the proposed research as a collaborative program rather than through separate research efforts? Will the research efforts taken together have more impact on the field than each separate project conducted in isolation? Is there strong synergy among the combined efforts of the various investigators within the overall Hub? Are the size and structure of the Hub sufficient to afford effective interaction focused on a specific central theme, but diverse enough in scientific expertise to achieve meaningful contributions to advancing data science and innovation?

Study Timeline

Specific to applications involving clinical trials:

Is the study timeline described in detail, taking into account start-up activities, the anticipated rate of enrollment, and planned follow-up assessment? Is the projected timeline feasible and well justified? Does the project incorporate efficiencies and utilize existing resources (e.g., practice-based research networks, electronic medical records, administrative database, or patient registries) to increase the efficiency of participant enrollment and data collection, as appropriate?

Are potential challenges and corresponding solutions discussed (e.g., strategies that can be implemented in the event of enrollment shortfalls)?

Protections for Human Subjects

For research that involves human subjects but does not involve one of the categories of research that are exempt under 45 CFR Part 46, the committee will evaluate the justification for involvement of human subjects and the proposed protections from research risk relating to their participation according to the following five review criteria: 1) risk to subjects, 2) adequacy of protection against risks, 3) potential benefits to the subjects and others, 4) importance of the knowledge to be gained, and 5) data and safety monitoring for clinical trials.

For research that involves human subjects and meets the criteria for one or more of the categories of research that are exempt under 45 CFR Part 46, the committee will evaluate: 1) the justification for the exemption, 2) human subjects involvement and characteristics, and 3) sources of materials. For additional information on review of the Human Subjects section, please refer to the [Guidelines for the Review of Human Subjects \(//grants.nih.gov/grants/guide/redirect.htm?id=11175\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11175).

Inclusion of Women, Minorities, and Individuals Across the Lifespan

When the proposed project involves human subjects and/or NIH-defined clinical research, the committee will evaluate the proposed plans for the inclusion (or exclusion) of individuals on the basis of sex/gender, race, and ethnicity, as well as the inclusion (or exclusion) of individuals of all ages (including children and older adults) to determine if it is justified in terms of the scientific goals and research strategy proposed. For additional information on review of the Inclusion section, please refer to the [Guidelines for the Review of Inclusion in Clinical Research \(//grants.nih.gov/grants/guide/redirect.htm?id=11174\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11174).

Vertebrate Animals

The committee will evaluate the involvement of live vertebrate animals as part of the scientific assessment according to the following criteria: (1) description of proposed procedures involving animals, including species, strains, ages, sex, and total number to be used; (2) justifications for the use of animals versus alternative models and for the appropriateness of the species proposed; (3) interventions to minimize discomfort, distress, pain and injury; and (4) justification for euthanasia method if NOT consistent with the AVMA Guidelines for the Euthanasia of Animals. Reviewers will assess the use of chimpanzees as they would any other application proposing the use of vertebrate animals. For additional information on review of the Vertebrate Animals section, please refer to the [Worksheet for Review of the Vertebrate Animal Section \(//grants.nih.gov/grants/guide/redirect.htm?id=11150\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11150).

Biohazards

Reviewers will assess whether materials or procedures proposed are potentially hazardous to research personnel and/or the environment, and if needed, determine whether adequate protection is proposed.

Resubmissions

Not Applicable

Renewals

Not Applicable

Revisions

Not Applicable

Additional Review Considerations - Overall

As applicable for the DS-I Africa Research Hub proposed, reviewers will consider each of the following items, but will not give scores for these items, and should not consider them in providing an overall impact score.

Applications from Foreign Organizations

Reviewers will assess whether the project presents special opportunities for furthering research programs through the use of unusual talent, resources, populations, or environmental conditions that exist in other countries and either are not readily available in the United States or augment existing U.S. resources.

Select Agent Research

Reviewers will assess the information provided in this section of the application, including 1) the Select Agent(s) to be used in the proposed research, 2) the registration status of all entities where Select Agent(s) will be used, 3) the procedures that will be used to monitor possession use and transfer of Select Agent(s), and 4) plans for appropriate biosafety, biocontainment, and security of the Select Agent(s).

Resource Sharing Plans

Reviewers will comment on whether the following Resource Sharing Plans, or the rationale for not sharing the following types of resources, are reasonable: 1) [Data Sharing Plan \(//grants.nih.gov/grants/guide/redirect.htm?id=11151\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11151); 2) [Sharing Model Organisms \(//grants.nih.gov/grants/guide/redirect.htm?id=11152\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11152); and 3) [Genomic Data Sharing Plan \(//grants.nih.gov/grants/guide/notice-files/NOT-OD-14-124.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-14-124.html). Reviewers will comment on whether the Summary of Hub Data Management and Analysis Plan is reasonable.

Authentication of Key Biological and/or Chemical Resources

For projects involving key biological and/or chemical resources, reviewers will comment on the brief plans proposed for identifying and ensuring the validity of those resources.

Budget and Period of Support

Reviewers will consider whether the budget and the requested period of support are fully justified and reasonable in relation to the proposed research.

Overall Impact Research Projects

In addition to the Overall Research Hub impact score and criterion scores, reviewers will provide an overall impact score and distinct criterion scores for each Research Project to reflect their assessment of the likelihood for the Project to exert a sustained, powerful influence on the research field(s) involved, in consideration of the following review criteria and additional review criteria (as applicable for the project proposed). These distinct Research Project impact scores will serve to 1) inform reviewers as they determine the Overall Research Hub impact score and 2) provide guidance to applicants for potentially addressing weaknesses in the proposed research plans prior to funding.

In addition, for applications involving clinical trials: A proposed Clinical Trial application may include study design, methods, and intervention that are not by themselves innovative but address important questions or unmet needs. Additionally, the results of the clinical trial may indicate that further clinical development of the intervention is unwarranted or lead to new avenues of scientific investigation.

Review Criteria Specific to Research Projects

Reviewers will provide an overall impact score to each project in consideration of the five Scored Review Criteria (Significance, Investigator(s), Innovation, Approach, and Environment) and its contribution to Overall DS-I Africa Research Hub (see Additional Review Criteria for all Projects and Cores). A separate score will be given for each of the five scored review criteria as part of the determination of scientific merit. A project does not need to be strong in all categories to be judged likely to have major scientific impact. For example, a Research Project that by its nature is not innovative may be essential to advance a field. The "Contribution to Overall Research Hub" criterion is not scored separately, but is considered in the determination of the overall impact score of the project.

Significance (Research Projects)

Does the project address an important problem or a critical barrier to progress in the field? Is the prior research that serves as the key support for the proposed project rigorous? If the aims of the project are achieved, how will scientific knowledge, technical capability, and/or clinical practice be improved? How will successful completion of the aims change the concepts, methods,

technologies, treatments, services, or preventative interventions that drive this field? To what extent do the goals of the proposed project contribute to the Research Hub overall? How will the research proposed be enhanced by being part of the Research Hub?

If the study is successful, would it lead to incremental advance, or would it provide a substantial step forward that would likely not be achieved through mechanisms other than this multi-project Research Hub? If successful, will the project result in knowledge or resources that could be utilized to improve human health and/or move field of data science forward, especially in the low-resource settings in Africa? If successful, would the project provide data, information, and knowledge to inform the processes of harnessing data science for health discovery and innovation in Africa? If applicable, will the project provide rigorous scientific data that might be used for effective decision-making by stakeholders?

In addition, for projects involving clinical trials:

Are the scientific rationale and need for a clinical trial to test the proposed hypothesis or intervention well supported by preliminary data, clinical and/or preclinical studies, or information in the literature or knowledge of biological mechanisms? For trials focusing on clinical or public health endpoints, is this clinical trial necessary for testing the safety, efficacy or effectiveness of an intervention that could lead to a change in clinical practice, community behaviors or health care policy? For trials focusing on mechanistic, behavioral, physiological, biochemical, or other biomedical endpoints, is this trial needed to advance scientific understanding?

Investigator(s) (Research Projects)

Are the Lead(s), collaborators, and other researchers well suited to the project? If Early Stage Investigators or in the early stages of independent careers, do they have appropriate experience and training? If established, have they demonstrated an ongoing record of accomplishments that have advanced their field(s)? If there is more than one Lead, do the investigators have complementary and integrated expertise? Do the investigators have a track record of success in carrying out data science-related or complementary research programs? Do the investigators demonstrate significant experience with coordinating collaborative research? If applicable, do the investigators have the experience and ability to handle diverse data types and to present an integrated view of these data?

In addition, for projects involving clinical trials:

With regard to the proposed leadership for the project, do the PD/PI(s) and key personnel have the expertise, experience, and ability to organize, manage and implement the proposed clinical trial and meet milestones and timelines? Do they have appropriate expertise in study coordination, data management and statistics?

Innovation (Research Projects)

Does the application challenge and seek to shift current research or clinical practice paradigms by utilizing novel theoretical concepts, approaches or methodologies, instrumentation, or interventions? Are the concepts, approaches or methodologies, instrumentation, or interventions novel to one field of research or novel in a broad sense? Is a refinement, improvement, or new application of theoretical concepts, approaches or methodologies, instrumentation, or interventions proposed? As applicable, does the project propose an innovative solution to a critical barrier to progress in the understanding of how data science and innovation can be leveraged to address critical health priorities in the African country or region of interest?

In addition, for projects involving clinical trials:

Does the design/research plan include innovative elements, as appropriate, that enhance its sensitivity, potential for information or potential to advance scientific knowledge or clinical practice?

Approach (Research Projects)

Are the overall strategy, methodology, and analyses well-reasoned and appropriate to accomplish the specific aims of the project? Have investigators included plans to address weaknesses in the rigor of prior research that serves as the key support for the proposed project? Have the investigators presented strategies to ensure a robust and unbiased approach, as appropriate for the work proposed? Are potential problems, alternative strategies, and benchmarks for success presented? If the project is in the early stages of development, will the strategy establish feasibility and will particularly risky aspects be managed? Have the investigators presented adequate plans to address relevant biological variables, such as sex, for studies in vertebrate animals or human subjects?

Does the project describe a strategy for stakeholder engagement with clear objectives of what they expect to achieve? Are there adequate plans to evaluate whether the strategies to engage the stakeholders and communities of interest are successful and the objectives achieved? Are the evaluation plans, milestones and timelines proposed appropriate and adequate for the project?

Has the issue of future sustainability been adequately addressed?

In addition, for applications involving clinical trials:

Does the application adequately address the following, if applicable:

Study Design

Is the study design justified and appropriate to address primary and secondary outcome variable(s)/endpoints that will be clear, informative and relevant to the hypothesis being tested? Is the scientific rationale/premise of the study based on previously well-designed preclinical and/or clinical research? Given the methods used to assign participants and deliver interventions, is the study design adequately powered to answer the research question(s), test the proposed hypothesis/hypotheses, and provide interpretable results? Is the trial appropriately designed to conduct the research efficiently? Are the study populations (size, gender, age, demographic group), proposed intervention arms/dose, and duration of the trial, appropriate and well justified?

Are potential ethical issues adequately addressed? Is the process for obtaining informed consent or assent appropriate? Is the eligible population available? Are the plans for recruitment outreach, enrollment, retention, handling dropouts, missed visits, and losses to follow-up appropriate to ensure robust data collection? Are the planned recruitment timelines feasible and is the plan to monitor accrual adequate? Has the need for randomization (or not), masking (if appropriate), controls, and inclusion/exclusion criteria been addressed? Are differences addressed, if applicable, in the intervention effect due to sex/gender and race/ethnicity?

Are the plans to standardize, assure quality of, and monitor adherence to, the trial protocol and data collection or distribution guidelines appropriate? Is there a plan to obtain required study agent(s)? Does the application propose to use existing available resources, as applicable?

Data Management and Statistical Analysis

Are planned analyses and statistical approach appropriate for the proposed study design and methods used to assign participants and deliver interventions? Are the procedures for data management and quality control of data adequate at clinical site(s) or at center laboratories, as applicable? Have the methods for standardization of procedures for data management to assess the effect of the intervention and quality control been addressed? Is there a plan to complete data analysis within the proposed period of the award?

Environment (Research Projects)

Will the scientific environment in which the work will be done contribute to the probability of success? Are the institutional support, equipment and other physical resources available to the investigators adequate for the project proposed? Will the project benefit from unique features of the scientific environment, subject populations, or collaborative arrangements?

In addition, for applications involving clinical trials:

If proposed, are the administrative, data coordinating, enrollment and laboratory/testing centers, appropriate for the trial proposed? Does the application adequately address the capability and ability to conduct the trial at the proposed site(s) or centers? Are the plans to add or drop enrollment centers, as needed, appropriate? If international site(s) is/are proposed, does the application adequately address the complexity of executing the clinical trial? If multi-sites/centers, is there evidence of the ability of the individual site or center to: (1) enroll the proposed numbers; (2) adhere to the protocol; (3) collect and transmit data in an accurate and timely fashion; and, (4) operate within the proposed organizational structure?

Additional Review Criteria - Research Projects and Cores

As applicable for the project proposed, reviewers will evaluate the following additional items while determining scientific and technical merit, and in providing an overall impact score, but will not give separate scores for these items.

Review Criteria Specific to Cores

Each Core will be reviewed based on its criteria below and its contribution to the overall DS-I Africa Research Hub (see "Additional Review Criteria - Research Projects and Cores" below). Each Core will not receive distinct impact or criterion scores but review of all Cores will inform the Overall Research Hub impact score.

Administrative Core

Are the objectives of the Administrative Core appropriate to the Hub, and do they reflect the interdisciplinary and multisectoral nature of the Research Hub? Is there evidence that the lines of authority and structure are designed for effective management/leadership of the Hub? Are the qualifications, duties, and time commitments of administrative staff (e.g., Hub Director, Center Administrator, etc.) appropriate? Does the Hub Director have appropriate experience and have they demonstrated effective and responsible leadership in the past? If there is more than one Lead, do the investigators have

complementary and integrated expertise? If applicable, is the role/expertise of the Deputy Director appropriate for the Administrative Core?

Are the institutional support, equipment and other physical resources available to the investigators adequate for the project proposed? Will the project benefit from unique features of the scientific environment or collaborative arrangements? Is there evidence of an internal plan to promote integration and coordination; data analysis/integration/sharing among the projects and cores; fiscal/resource/quality management oversight; and interactions between projects, cores, and external partners/communities? Is the plan for reporting updates to relevant stakeholders (including, but not limited to, NIH staff, DS-I Africa Coordinating Center, DS-I Africa Open Data Science Platform, and other DS-I Africa Consortium component organizations) adequate? Are plans for partnerships with government agencies and businesses, as well as information dissemination to other end-users adequate? Is there a plan for coordinating with the Data Management and Analysis Core and optional Support Cores? Does the Core promote and/or enhance interactions among the Hub components?

Does the Administrative Core provide an adequate plan to solicit, select, and evaluate promising data science-related pilot projects for sub-award funding that have significant potential to address developing needs, from both within and outside the Hub institution? Does the core's strategy for selecting pilot projects suggest potential for meaningful outcomes in the chosen area of data science research or clinical application? Will the core's evaluation processes enable selection of promising pilot projects for funding and robust monitoring and evaluation of project progress and implementation? Is there an appropriate plan to establish/utilize a scientific advisory board?

Data Management and Analysis Core

Are the objectives of the Data Management and Analysis Core (DMAC) appropriate to the Research Hub? Is the Comprehensive Data Management Plan adequate and does it include plans for coordination with projects and cores; fostering data sharing and interoperability; and data quality assurance and quality control? Are the approaches, methods, and expertise proposed adequate to achieve objectives? If the aims of the DMAC are achieved, how will scientific knowledge, technical capability, and/or clinical practice be improved? Do the investigators have the experience and ability to handle diverse data types and to present an integrated view of these data? Does the DMAC promote and/or enhance interactions among the Hub components? Does the DMAC have a reasonable plan to prioritize datasets to be shared; to identify platforms for sharing; and to establish appropriate timelines for dataset deposition? Are the plans to support the data analytic, integration, and sharing efforts of projects and cores reasonable and adequate? For Hubs proposing additional functionality within the DMAC (e.g., biostatistics, GIS, etc.), is expertise adequate to support additional functionality? Are applicable plans for technology transfer and/or intellectual property management well developed and appropriate for potentially scaling innovations developed within the Research Hub? Does the additional functionality enhance or detract from the data management and analysis goals of the DMAC? If provided, is the example blank data management plan template(s) used for the Hub's projects/cores adequate?

Support Core(s) (optional)

Are the objectives of the optional Support Core(s) appropriate to the Hub? Are the approaches and methods proposed adequate to achieve objectives? Does the Support Core promote and/or enhance interactions among the Hub components? Does each Support Core provide essential facilities or service for the Research Projects and Pilot Projects? Is the projected use sufficient to warrant establishment of the core? Are the core facilities contributing to the overall research activities of the Hub? Is there evidence of enhanced efficiencies (including cost) afforded by the core? Is the Quality Assurance Statement for cores providing quantitative analyses adequate? Is there a prioritization plan for use of core facilities/services? Does the staff have the appropriate experience and level of commitment?

Contribution to Overall Research Hub (applies to all Projects and Cores)

Does the project/core contribute scientifically to the proposed Hub's problem-solving goals (e.g., the importance of the ideas or aims, the rationale and originality of the approach, the feasibility of the methods, and the value of the result)? Will the specific scientific objectives of each project/core benefit significantly from, or depend upon, collaborative interactions with other projects in the Hub (e.g., objectives that can be uniquely accomplished, specific contributions to the accomplishments of objectives in other projects, and objectives that can be accomplished with greater effectiveness and/or economy of effort)?

Study Timeline

Specific to applications involving clinical trials:

Is the study timeline described in detail, taking into account start-up activities, the anticipated rate of enrollment, and planned follow-up assessment? Is the projected timeline feasible and well justified? Does the project incorporate efficiencies and utilize

existing resources (e.g., practice-based research networks, electronic medical records, administrative database, or patient registries) to increase the efficiency of participant enrollment and data collection, as appropriate?

Are potential challenges and corresponding solutions discussed (e.g., strategies that can be implemented in the event of enrollment shortfalls)?

Protections for Human Subjects

For research that involves human subjects but does not involve one of the categories of research that are exempt under 45 CFR Part 46, the committee will evaluate the justification for involvement of human subjects and the proposed protections from research risk relating to their participation according to the following five review criteria: 1) risk to subjects, 2) adequacy of protection against risks, 3) potential benefits to the subjects and others, 4) importance of the knowledge to be gained, and 5) data and safety monitoring for clinical trials.

For research that involves human subjects and meets the criteria for one or more of the categories of research that are exempt under 45 CFR Part 46, the committee will evaluate: 1) the justification for the exemption, 2) human subjects involvement and characteristics, and 3) sources of materials. For additional information on review of the Human Subjects section, please refer to the [Guidelines for the Review of Human Subjects \(//grants.nih.gov/grants/guide/redirect.htm?id=11175\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11175).

Inclusion of Women, Minorities, and Individuals Across the Lifespan

When the proposed research project involves human subjects and/or NIH-defined clinical research, the committee will evaluate the proposed plans for the inclusion (or exclusion) of individuals on the basis of sex/gender, race, and ethnicity, as well as the inclusion (or exclusion) of individuals of all ages (including children and older adults) to determine if it is justified in terms of the scientific goals and research strategy proposed. For additional information on review of the Inclusion section, please refer to the [Guidelines for the Review of Inclusion in Clinical Research \(//grants.nih.gov/grants/guide/redirect.htm?id=11174\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11174).

Vertebrate Animals

The committee will evaluate the involvement of live vertebrate animals as part of the scientific assessment according to the following criteria: (1) description of proposed procedures involving animals, including species, strains, ages, sex, and total number to be used; (2) justifications for the use of animals versus alternative models and for the appropriateness of the species proposed; (3) interventions to minimize discomfort, distress, pain and injury; and (4) justification for euthanasia method if NOT consistent with the AVMA Guidelines for the Euthanasia of Animals. Reviewers will assess the use of chimpanzees as they would any other application proposing the use of vertebrate animals. For additional information on review of the Vertebrate Animals section, please refer to the [Worksheet for Review of the Vertebrate Animal Section \(//grants.nih.gov/grants/guide/redirect.htm?id=11150\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11150).

Biohazards

Reviewers will assess whether materials or procedures proposed are potentially hazardous to research personnel and/or the environment, and if needed, determine whether adequate protection is proposed.

Resubmissions

Not Applicable

Renewals

Not Applicable

Revisions

Not Applicable

Additional Review Considerations Research Projects and Cores

As applicable for the project/core proposed, reviewers will consider each of the following items, but will not give scores for these items, and should not consider them in providing an overall impact/priority score.

Select Agent Research

Reviewers will assess the information provided in this section of the application, including 1) the Select Agent(s) to be used in the proposed research, 2) the registration status of all entities where Select Agent(s) will be used, 3) the procedures that will be used

to monitor possession use and transfer of Select Agent(s), and 4) plans for appropriate biosafety, biocontainment, and security of the Select Agent(s).

Resource Sharing Plans

Reviewers will comment on whether the following Resource Sharing Plans, or the rationale for not sharing the following types of resources, are reasonable: 1) [Data Sharing Plan \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=11151\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11151); 2) [Sharing Model Organisms \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=11152\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11152); and 3) [Genomic Data Sharing Plan \(https://grants.nih.gov/grants/guide/notice-files/NOT-OD-14-124.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-14-124.html). For research projects, the reviewers will comment on whether the "Data Management Plan" is reasonable.

Authentication of Key Biological and/or Chemical Resources

For projects involving key biological and/or chemical resources, reviewers will comment on the brief plans proposed for identifying and ensuring the validity of those resources.

Budget and Period of Support

Reviewers will consider whether the budget and the requested period of support are fully justified and reasonable in relation to the proposed research.

2. Review and Selection Process

Applications will be evaluated for scientific and technical merit by (an) appropriate Scientific Review Group(s), convened by the Center for Scientific Review in accordance with [NIH peer review policy and procedures \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=11154\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11154), using the stated [review criteria](#). Assignment to a Scientific Review Group will be shown in the eRA Commons.

As part of the scientific peer review, all applications:

- May undergo a selection process in which only those applications deemed to have the highest scientific and technical merit (generally the top half of applications under review) will be discussed and assigned an overall impact score.
- Will receive a written critique.
- [Appeals \(https://grants.nih.gov/grants/guide/notice-files/NOT-OD-11-064.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-11-064.html) of initial peer review will not be accepted for applications submitted in response to this FOA.

Applications will be assigned to the appropriate NIH Institute or Center. Applications will compete for available funds with all other recommended applications submitted in response to this FOA. Following initial peer review, recommended applications will receive a second level of review by the Fogarty International Center Advisory Board. The following will be considered in making funding decisions:

- Scientific and technical merit of the proposed project as determined by scientific peer review.
- Availability of funds.
- Relevance of the proposed project to program priorities.
- Relevance of the proposed project to the goals of DS-I Africa.
- Relevance of proposed project to the priorities of co-funding components.
- Commitment to compliance with DS-I Africa procedures.
- Consideration of programmatic, geographic, and institutional distribution.

3. Anticipated Announcement and Award Dates

After the peer review of the application is completed, the PD/PI will be able to access his or her Summary Statement (written critique) via the [eRA Commons \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=11123\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11123). Refer to Part 1 for dates for peer review, advisory council review, and earliest start date.

Information regarding the disposition of applications is available in the [NIH Grants Policy Statement \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=11156\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11156).

Section VI. Award Administration Information

1. Award Notices

If the application is under consideration for funding, NIH will request "just-in-time" information from the applicant as described in the [NIH Grants Policy Statement \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=11157\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11157).

A formal notification in the form of a Notice of Award (NoA) will be provided to the applicant organization for successful applications. The NoA signed by the grants management officer is the authorizing document and will be sent via email to the grantee's business official.

Awardees must comply with any funding restrictions described in [Section IV.5. Funding Restrictions](#). Selection of an application for award is not an authorization to begin performance. Any costs incurred before receipt of the NoA are at the recipient's risk. These costs may be reimbursed only to the extent considered allowable pre-award costs.

Any application awarded in response to this FOA will be subject to terms and conditions found on the [Award Conditions and Information for NIH Grants](#) (https://grants.nih.gov/grants/guide/url_redirect.htm?id=11158) website. This includes any recent legislation and policy applicable to awards that is highlighted on this website.

Institutional Review Board or Independent Ethics Committee Approval: Grantee institutions must ensure that protocols are reviewed by their IRB or IEC. To help ensure the safety of participants enrolled in NIH-funded studies, the awardee must provide NIH copies of documents related to all major changes in the status of ongoing protocols.

Individual awards are based on the application submitted to, and as approved by, the NIH and are subject to the IC-specific terms and conditions identified in the NoA.

ClinicalTrials.gov: If an award provides for one or more clinical trials. By law (Title VIII, Section 801 of Public Law 110-85), the "responsible party" must register and submit results information for certain applicable clinical trials on the ClinicalTrials.gov Protocol Registration and Results System Information Website (<https://register.clinicaltrials.gov> (<https://register.clinicaltrials.gov>)). NIH expects registration and results reporting of all trials whether required under the law or not. For more information, see <https://grants.nih.gov/policy/clinical-trials/reporting/index.htm> (<https://grants.nih.gov/policy/clinical-trials/reporting/index.htm>).

Data and Safety Monitoring Requirements: The NIH policy for data and safety monitoring requires oversight and monitoring of all NIH-conducted or -supported human biomedical and behavioral intervention studies (clinical trials) to ensure the safety of participants and the validity and integrity of the data. Further information concerning these requirements is found at http://grants.nih.gov/grants/policy/hs/data_safety.htm and in the application instructions (SF424 (R&R) and PHS 398).

Investigational New Drug or Investigational Device Exemption Requirements: Consistent with federal regulations, clinical research projects involving the use of investigational therapeutics, vaccines, or other medical interventions (including licensed products and devices for a purpose other than that for which they were licensed) in humans under a research protocol must be performed under a Food and Drug Administration (FDA) investigational new drug (IND) or investigational device exemption (IDE).

Prior Approval of Pilot Projects

Awardee-selected projects that involve {clinical trials or studies involving greater than minimal risk to human subjects} require prior approval by NIH prior to initiation.

- The awardee institution will comply with the NIH [Guidance on Changes That Involve Human Subjects in Active Awards and That Will Require Prior NIH Approval](#) (<https://grants.nih.gov/grants/guide/notice-files/NOT-OD-15-128.html>).
- The awardee institution will provide NIH with specific plans for data and safety monitoring, and will notify the IRB and NIH of serious adverse events and unanticipated problems, consistent with [NIH DSMP policies](#) (https://grants.nih.gov/grants/guide/url_redirect.htm?id=21600).
- See Cooperative Agreement Terms and Conditions of Award for further information regarding joint PD/PI(s) and NIH staff involvement in approval of pilot projects.

2. Administrative and National Policy Requirements

All NIH grant and cooperative agreement awards include the [NIH Grants Policy Statement](#) (https://grants.nih.gov/grants/guide/url_redirect.htm?id=11120) as part of the NoA. For these terms of award, see the [NIH Grants Policy Statement Part II: Terms and Conditions of NIH Grant Awards, Subpart A: General](#) (https://grants.nih.gov/grants/guide/url_redirect.htm?id=11157) and [Part II: Terms and Conditions of NIH Grant Awards, Subpart B: Terms and Conditions for Specific Types of Grants, Grantees, and Activities](#) (https://grants.nih.gov/grants/guide/url_redirect.htm?id=11159). More information is provided at [Award Conditions and Information for NIH Grants](#) (https://grants.nih.gov/grants/guide/url_redirect.htm?id=11158).

Recipients of federal financial assistance (FFA) from HHS must administer their programs in compliance with federal civil rights laws that prohibit discrimination on the basis of race, color, national origin, disability, age and, in some circumstances, religion, conscience, and sex. This includes ensuring programs are accessible to persons with limited English proficiency. The HHS Office for Civil Rights provides guidance on complying with civil rights laws enforced by HHS. Please see <https://www.hhs.gov/civil-rights/for-providers/provider-obligations/index.html> (<https://www.hhs.gov/civil-rights/for-providers/provider-obligations/index.html>) and

<http://www.hhs.gov/ocr/civilrights/understanding/section1557/index.html>
(<https://www.hhs.gov/ocr/civilrights/understanding/section1557/index.html>).

HHS recognizes that research projects are often limited in scope for many reasons that are nondiscriminatory, such as the principal investigator's scientific interest, funding limitations, recruitment requirements, and other considerations. Thus, criteria in research protocols that target or exclude certain populations are warranted where nondiscriminatory justifications establish that such criteria are appropriate with respect to the health or safety of the subjects, the scientific study design, or the purpose of the research. For additional guidance regarding how the provisions apply to NIH grant programs, please contact the Scientific/Research Contact that is identified in Section VII under Agency Contacts of this FOA.

- Recipients of FFA must ensure that their programs are accessible to persons with limited English proficiency. HHS provides guidance to recipients of FFA on meeting their legal obligation to take reasonable steps to provide meaningful access to their programs by persons with limited English proficiency. Please see <https://www.hhs.gov/civil-rights/for-individuals/special-topics/limited-english-proficiency/fact-sheet-guidance/index.html> (<https://www.hhs.gov/civil-rights/for-individuals/special-topics/limited-english-proficiency/fact-sheet-guidance/index.html>) and <https://www.lep.gov> (<https://www.lep.gov>). For further guidance on providing culturally and linguistically appropriate services, recipients should review the National Standards for Culturally and Linguistically Appropriate Services in Health and Health Care at <https://minorityhealth.hhs.gov/omh/browse.aspx?lvl=2&lvlid=53> (<https://minorityhealth.hhs.gov/omh/browse.aspx?lvl=2&lvlid=53>).
- Recipients of FFA also have specific legal obligations for serving qualified individuals with disabilities. Please see <http://www.hhs.gov/ocr/civilrights/understanding/disability/index.html> (<https://www.hhs.gov/ocr/civilrights/understanding/disability/index.html>).
- HHS funded health and education programs must be administered in an environment free of sexual harassment. Please see <https://www.hhs.gov/civil-rights/for-individuals/sex-discrimination/index.html> (<https://www.hhs.gov/civil-rights/for-individuals/sex-discrimination/index.html>); <https://www2.ed.gov/about/offices/list/ocr/docs/shguide.html>; and <https://www.eeoc.gov/eeoc/publications/upload/fs-sex.pdf> (<https://www.eeoc.gov/eeoc/publications/upload/fs-sex.pdf>). For information about NIH's commitment to supporting a safe and respectful work environment, who to contact with questions or concerns, and what NIH's expectations are for institutions and the individuals supported on NIH-funded awards, please see <https://grants.nih.gov/grants/policy/harassment.htm> (<https://grants.nih.gov/grants/policy/harassment.htm>).
- Recipients of FFA must also administer their programs in compliance with applicable federal religious nondiscrimination laws and applicable federal conscience protection and associated anti-discrimination laws. Collectively, these laws prohibit exclusion, adverse treatment, coercion, or other discrimination against persons or entities on the basis of their consciences, religious beliefs, or moral convictions. Please see <https://www.hhs.gov/conscience/conscience-protections/index.html> (<https://www.hhs.gov/conscience/conscience-protections/index.html>) and <https://www.hhs.gov/conscience/religious-freedom/index.html> (<https://www.hhs.gov/conscience/religious-freedom/index.html>).

Please contact the HHS Office for Civil Rights for more information about obligations and prohibitions under federal civil rights laws at <https://www.hhs.gov/ocr/about-us/contact-us/index.html> (<https://www.hhs.gov/ocr/about-us/contact-us/index.html>) or call 1-800-368-1019 or TDD 1-800-537-7697.

In accordance with the statutory provisions contained in Section 872 of the Duncan Hunter National Defense Authorization Act of Fiscal Year 2009 (Public Law 110-417), NIH awards will be subject to the Federal Awardee Performance and Integrity Information System (FAPIS) requirements. FAPIS requires Federal award making officials to review and consider information about an applicant in the designated integrity and performance system (currently FAPIS) prior to making an award. An applicant, at its option, may review information in the designated integrity and performance systems accessible through FAPIS and comment on any information about itself that a Federal agency previously entered and is currently in FAPIS. The Federal awarding agency will consider any comments by the applicant, in addition to other information in FAPIS, in making a judgement about the applicant's integrity, business ethics, and record of performance under Federal awards when completing the review of risk posed by applicants as described in 45 CFR Part 75.205 Federal awarding agency review of risk posed by applicants. This provision will apply to all NIH grants and cooperative agreements except fellowships.

Cooperative Agreement Terms and Conditions of Award

The following special terms of award are in addition to, and not in lieu of, otherwise applicable U.S. Office of Management and Budget (OMB) administrative guidelines, U.S. Department of Health and Human Services (DHHS) grant administration regulations at 45 CFR Part 75, and other HHS, PHS, and NIH grant administration policies.

The administrative and funding instrument used for this program will be the cooperative agreement, an "assistance" mechanism (rather than an "acquisition" mechanism), in which substantial NIH programmatic involvement with the awardees is anticipated during the performance of the activities. Under the cooperative agreement, the NIH purpose is to support and stimulate the recipients'

activities by involvement in and otherwise working jointly with the award recipients in a partnership role; it is not to assume direction, prime responsibility, or a dominant role in the activities. Consistent with this concept, the dominant role and prime responsibility resides with the awardees for the project as a whole, although specific tasks and activities may be shared among the awardees and the NIH as defined below.

Awardee and Principal Investigator Primary Responsibilities

- Planning, directing, and executing all aspects of the proposed research activities, including any modification of study design, conduct of the study, quality control, data analysis and interpretation, preparation of publications, dissemination of data, tools, and technologies, and collaboration with other investigators, unless otherwise provided for in these terms;
- Providing effective leadership and management of all activities under the award;
- Developing programmatic goals, timelines, and milestones, reporting of progress to NIH biannually, and ensuring that the outputs produced meets the quality standards agreed to at the beginning of the project by the DS-I Africa Consortium;
- Assuring compliance with all applicable DHHS/NIH policies for conduct of research and clinical trials as well as host country regulations for protection of human subjects and conduct of clinical research;
- Ensuring collaboration with the DS-I Africa Open Data Science Platform, that data resources developed as part of this project are shared according to DS-I Africa policies, and that results are published in a timely manner;
- Establishing and managing a solicitation, review, and selection process for collaborative sub-award pilot projects aligned with the Research Hub overall focus and milestones, including the distribution of yearly sub-award funds;
- Working in close coordination, cooperation, and participation with NIH staff for scientific and technical management of the project, described further under NIH Responsibilities;
- Agreeing to the governance of the Consortium through the Steering Committee, including adherence to DS-I Africa policies and procedures; and
- Serving as active members of the DS-I Africa Steering Committee, attending Steering Committee meetings, and participating directly or by proxy on relevant DS-I Africa Working Groups, both remotely through regular teleconferences and in-person at biannual DS-I Africa Consortium Meetings.

Awardees will retain custody of and have primary rights to the data and software developed under these awards, subject to Government rights of access consistent with current DHHS, PHS, and NIH policies.

NIH Responsibilities

NIH staff (including, but not limited to, Project Scientists, Program Officers, Intramural Scientists, and Program Analysts) have substantial programmatic involvement that is above and beyond the normal stewardship role in awards.

NIH Project Scientists roles include:

- Participating (with DS-I Africa investigators) in the group process of deciding optimal research approaches and protocol designs, and contributing to the adjustment of research protocols or approaches as warranted;
- Attending Steering Committee meetings, serving on DS-I Africa Working Groups, and assisting in developing operating guidelines, quality control procedures, and consistent policies for dealing with recurrent situations that require coordinated action;

NIH Program Officer roles include:

- Negotiating goals and milestones with the awardees as needed;
- Serving as liaisons between the awardees and the appropriate NIH Institute and Center National Advisory Councils and the NIH DS-I Africa Working Group made up of program staff that collectively manage the program;
- Attending Steering Committee meetings, serving on DS-I Africa Working Groups, and assisting in developing operating guidelines, quality control procedures, and consistent policies for dealing with recurrent situations that require coordinated action;
- Convening external consultants to support NIH oversight of the DS-I Africa consortium?;
- Interacting with the PD(s)/PI(s) on a regular basis to monitor progress, which may include regular communication with the study leadership and staff, periodic site visits for discussion with the awardee research team, observation of data collection and management techniques, fiscal reviews, and other relevant stewardship matters; and
- Making recommendations for continued funding based on (a) overall research progress; (b) cooperation in carrying out the research (e.g., attendance at Steering Committee meetings, implementation of group decisions, compliance with the terms of award and reporting requirements); and/or (c) maintenance of high-quality research.
- Normal scientific and programmatic stewardship of the award and assignment in the award notice. The assigned NIH Program Official may not also serve as an NIH Project Scientist.

The government, via NIH Program Scientists, will have access to all data generated under this Cooperative Agreement and may periodically review and analyze the data for the preparation of internal reports related to the Cooperative Agreement.

An NIH intramural scientist may not serve as the PD/PI of a project awarded under this FOA but may participate as a collaborator or consultant. For applications that include NIH intramural components, the involvement of intramural scientists needs to be consistent with NIH policy ?-? <https://oir.nih.gov/sourcebook/ethical-conduct/research-ethics/nih-policies/intramural-extramural-collaborations> (<https://oir.nih.gov/sourcebook/ethical-conduct/research-ethics/nih-policies/intramural-extramural-collaborations>).

The NIH reserves the right to withhold funding or curtail studies in an individual award in the event of (a) substantive changes in the agreed-upon work scope with which NIH cannot concur, (b) human subject ethical issues that may dictate a premature termination; (c) or insufficient project progress.

Collaborative Responsibilities

A Steering Committee (SC) will serve as the governing board for awardees. All participants in the program are bound by the policies and procedures developed by the SC and adoption of such policies and procedures requires a majority vote. Membership on the SC will include the PI(s) of each award and associated NIH program officers and projects scientists. On voting matters, each funded project will have one vote for determining appropriate collaborative activities, with NIH providing final approval for consortium activities. The chair will be chosen by a majority vote of the SC, with years of service as chair determined by the committee. The chair, in collaboration with the associated program Coordinating Center staff, is responsible for preparing meeting agendas, for chairing meetings, and for approving minutes in a timely manner. NIH staff may not serve as Chair of the SC.

Steering Committee responsibilities include:

- Discussing progress in meeting the broad goals of various DS-I Africa projects (while maintaining appropriate project-specific protection of intellectual property) and of DS-I Africa as a whole;
- Developing recommendations for uniform procedures and policies necessary to meet the goals of the Consortium, for example for data quality measures and assessment as well as conventions for data sharing;
- Cooperating to ensure the timely and broad dissemination of lessons learned to inform researchers and health care systems engaged in data science for health discovery and innovation in Africa;
- Endorsing and overseeing progress and products of Working Groups within the Consortium; and
- Meeting twice a year in-person in conjunction with Consortium meetings and conducting regularly scheduled conference calls.

Dispute Resolution

Any disagreements that may arise in scientific or programmatic matters (within the scope of the award) between award recipients and the NIH may be brought to Dispute Resolution and a Dispute Resolution Panel will be convened. The Panel will have three members: a designee of the Steering Committee chosen without NIH staff voting, one NIH designee, and a third designee with expertise in the relevant area who is chosen by the other two; in the case of individual disagreement, the first member may be chosen by the individual awardee. This special dispute resolution procedure does not alter the awardee's right to appeal an adverse action that is otherwise appealable in accordance with PHS regulation 42 CFR Part 50, Subpart D and DHHS regulation 45 CFR Part 16.

3. Reporting

When multiple years are involved, awardees will be required to submit the [Research Performance Progress Report \(RPPR\)](https://grants.nih.gov/grants/rppr/index.htm) (<https://grants.nih.gov/grants/rppr/index.htm>) annually and financial statements as required in the [NIH Grants Policy Statement](https://grants.nih.gov/grants/guide/redirect.htm?id=11161). (<https://grants.nih.gov/grants/guide/redirect.htm?id=11161>) Progress reports should briefly describe status of pilot projects, including data and safety monitoring, and should notify NIH of serious adverse events and unanticipated problems.

A final RPPR, invention statement, and the expenditure data portion of the Federal Financial Report are required for closeout of an award, as described in the [NIH Grants Policy Statement](https://grants.nih.gov/grants/guide/redirect.htm?id=11161) (<https://grants.nih.gov/grants/guide/redirect.htm?id=11161>).

The Federal Funding Accountability and Transparency Act of 2006 (Transparency Act), includes a requirement for awardees of Federal grants to report information about first-tier subawards and executive compensation under Federal assistance awards issued in FY2011 or later. All awardees of applicable NIH grants and cooperative agreements are required to report to the Federal Subaward Reporting System (FSRS) available at www.fsrs.gov (<https://grants.nih.gov/grants/guide/redirect.htm?id=11170>) on all subawards over \$25,000. See the [NIH Grants Policy Statement](https://grants.nih.gov/grants/guide/redirect.htm?id=11171) (<https://grants.nih.gov/grants/guide/redirect.htm?id=11171>) for additional information on this reporting requirement.

In accordance with the regulatory requirements provided at 45 CFR 75.113 and Appendix XII to 45 CFR Part 75, recipients that have currently active Federal grants, cooperative agreements, and procurement contracts from all Federal awarding agencies with a cumulative total value greater than \$10,000,000 for any period of time during the period of performance of a Federal award, must report and maintain the currency of information reported in the System for Award Management (SAM) about civil, criminal, and

administrative proceedings in connection with the award or performance of a Federal award that reached final disposition within the most recent five-year period. The recipient must also make semiannual disclosures regarding such proceedings. Proceedings information will be made publicly available in the designated integrity and performance system (currently FAPIIS). This is a statutory requirement under section 872 of Public Law 110-417, as amended (41 U.S.C. 2313). As required by section 3010 of Public Law 111-212, all information posted in the designated integrity and performance system on or after April 15, 2011, except past performance reviews required for Federal procurement contracts, will be publicly available. Full reporting requirements and procedures are found in Appendix XII to 45 CFR Part 75 Award Term and Conditions for Recipient Integrity and Performance Matters.

Section VII. Agency Contacts

We encourage inquiries concerning this funding opportunity and welcome the opportunity to answer questions from potential applicants.

Application Submission Contacts

eRA Service Desk (Questions regarding ASSIST, eRA Commons, application errors and warnings, documenting system problems that threaten submission by the due date, and post-submission issues)

Finding Help Online: <http://grants.nih.gov/support/> (preferred method of contact)

Telephone: 301-402-7469 or 866-504-9552 (Toll Free)

General Grants Information (Questions regarding application instructions, application processes, and NIH grant resources)

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Section VIII. Other Information

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